[Web Appendices]

A Topic-Based Segmentation Model for Identifying Segment-Level Drivers of Star Ratings from Unstructured Text Reviews

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These materials have been supplied by the authors to aid in the understanding of their paper. The AMA is sharing these materials at the request of the authors.

Web Appendix A: Preprocessing Steps to Clean the Unstructured Texts

Step 0. Import the unstructred text data.

Step 1. Change all cases to lower cases.

Example R code: tm_map(textdata, content_transformer(tolower)) # tm package in R

Step 2. Remove all punctuations.

Example code: tm_map(textdata, removePunctuation) # tm package

Step 3. Remove any stopwords (meaningless words). note, some specific words can be added to the list of stopwords by model users.

Example code: tm_map(textdata, removeWords, stopwords("english")) # tm package

Step 4. Remove additionally unncessary words that frequently appear, if there are any.

Example code: tm_map(textdata, removeWords, c("does","not","thing","unnecessary"))

Step 5. Remove extra whitespace from text documents.

Example code: tm_map(textdata, stripWhitespace) # tm package

Step 6. Remove numbers, if numbers are unnecessary. # tm package

Example code: tm_map(textdata, removeNumbers) # tm package

Step 7. Check the possible effect of typos.

a. Check the number (or proportion) of typos in the target text data

Example code: Correct <- hunspell_check(textdata) # hunspell package in R

b. After checking proportion of the typos, we can remove the typos.

Example code: Correct <- hunspell_check(textdata) # hunspell package

Index_typos <- which(correct==FALSE)

Note: As we checked our data with the code above, the proportion of typos in our application studies was about 3%. After confirming the proportion of the typos is small, we removed them.

c. If the proportion of the typos is somewhat large and there are any frequent, repetitive typos, model users can easily replace such typos by correct words.

Example code: gsub('typo', 'correctone', text)

Step 8. Transform all cleaned words to stemming words.

Example code: tm_map(textdata, stemDocument, language="english") # SnowballC package in R

Step 9. Build a word-document matrix (documents as row and words as column).

Example code: TermDocumentMatrix(textdata, control = list(removePunctuation = TRUE, stopwords = TRUE)) # tm package

Step 10. If there can be still too many words, we can remove some very sparse words.

Example code in R: removeSparseTerms(wordmatrix, 0.99) # tm package

Web Appendix B: Full Conditional Distributions for A Topic-Based Bayesian Finite Mixture Regression for Unstructured Review Texts

(1) Full conditional distribution of H_i:

Since

$$\pi(H_i = k|all\ others) \propto f\left(Y_i \middle| H_i = k, \underline{\beta}_k\right)\pi(H_i = k|d_k)$$

where $f\left(Y_i \middle| H_i = k, \underline{\beta}_k\right) \propto \exp\left(-\frac{(Y_i - \underline{X}_i^T \underline{\beta}_k)^2}{2\sigma^2}\right)$, we have

$$\pi(H_i = k|all\ others) = d_k^*$$

where $d_k^* = \frac{\exp\left(-\frac{(Y_i - \underline{X}_i^T \underline{\beta}_k)^2}{2\sigma^2}\right) d_k}{\sum_{l=1}^{K} \exp\left(-\frac{(Y_i - \underline{X}_i^T \underline{\beta}_l)^2}{2\sigma^2}\right) d_l}$. Thus,

$$\pi(H_i|all\ others) = Multinoulli(d_1^*, \dots, d_K^*)$$

(2) Full conditional distribution of $\underline{d}$:

$$\pi\left(\underline{d} \middle| all\ others\right) \propto \pi(\underline{H}|\underline{d})\pi(\underline{d})$$

$$\propto \prod_{k=1}^{K} {d_k}^{\alpha_k + n_k - 1}.$$

where $n_k = \sum_{i=1}^{n} \mathbf{1}_{\{H_i = k\}}$. Thus,

$$\pi\left(\underline{d} \middle| all\ others\right) = Dirichlet(\alpha_1 + n_1, \dots, \alpha_K + n_K).$$

(3) Full conditional distribution of β_{kp}:

If Z_{kp}=0,

$\pi\left(\beta_{kp} = 0 \middle| Z_{kp} = 0\ and\ others\right) = 1$,

If Z_{kp}=1,

$$\pi\left(\beta_{kp} \middle| Z_{kp} = 1\ and\ others\right) \propto \phi(\beta_{kp}|0, \tau_p^2) \prod_{i \epsilon i d_k} \phi(Y_i|\underline{X}_i^T \underline{\beta}_k, \sigma^2)$$

$$\propto e^{-\frac{\beta_{kp}^2}{2\tau_p^2}} e^{-\frac{\sum_{i\epsilon id_k}\left(Y_i-\underline{X}_{i[-p]}^T\underline{\beta}_{k[-p]}-X_{ip}\beta_{kp}\right)^2}{2\sigma^2}},$$

where $id_k = \{i: H_i = k\}$, $\underline{\mathrm{X}}_{\mathrm{i}[-p]}^{\mathrm{T}}$ is $\underline{\mathrm{X}}_{\mathrm{i}}^{\mathrm{T}}$ without the p[th] component, and $\underline{\beta}_{k[-p]}$ is $\underline{\beta}_k$ without its p[th] component. Thus,

$$\pi(\beta_{kp}|Z_{kp} = 1\ and\ others) = N\left(\frac{\frac{1}{\sigma^2}\sum_{i\epsilon id_k}\left(Y_i - \underline{\mathrm{X}}_{\mathrm{i}[-p]}^{\mathrm{T}}\underline{\beta}_{k[-p]}\right)X_{ip}}{\frac{1}{\tau_p^2}+\frac{1}{\sigma^2}\sum_{i\epsilon id_k}X_{ip}^2}, 1/(\frac{1}{\tau_p^2}+\frac{1}{\sigma^2}\sum_{i\epsilon id_k}X_{ip}^2)\right).$$

(4) Full conditional distribution of Z_{kp}:

Since Z_{kp} can take on two possible values, 0 and 1 only, so if we let:

$$R=\frac{\pi(Z_{kp}=0|all\ others\ except\ \beta_{kp})}{\pi(Z_{kp}=1|all\ others\ except\ \beta_{kp})},$$

then $\pi(Z_{kp}=0|all\ others\ except\ \beta_{kp}) = \frac{R}{1+R}$ and

$\pi(Z_{kp}=1|all\ others\ except\ \beta_{kp}) = \frac{1}{1+R}$,

or equivalently, $\pi(Z_{kp}|all\ others\ except\ \beta_{kp}) = Bernoulli\left(\frac{1}{1+R}\right)$.

To compute R, note that $\pi(Z_{kp}|all\ others\ except\ \beta_{kp}) = \pi(\beta_{kp}, Z_{kp}|all\ others)/$ $\pi(\beta_{kp}|all\ others)$,

and:

$$\pi(\beta_{kp}, Z_{kp}|all\ others) \propto \prod_{i\epsilon id_k}\phi\left(Y_i\left|\underline{X}_i^T\underline{\beta}_k, \sigma^2\right.\right)$$

$$\times\left[Z_{kp}\phi(\beta_{kp}|0,\tau_p^2) + (1-Z_{kp})\mathbf{1}_{\{\beta_{kp}=0\}}\right]\times\left[\exp\left(\lambda\sum_{p\sim\tilde{p}|k}w_{p\tilde{p}}\,I(z_{k\tilde{p}}=1)\right)\right]^{Z_{kp}},$$

here $(p\sim\tilde{p})|k$ indicates that words p and $\tilde{p}$ are neighbors in the k[th] segment. $w_{p\tilde{p}}$ is weight of the inverse of the min-max normalized distance between p and $\tilde{p}$ (nearer

distance, higher weight), $I(z_{k\tilde{p}} = 1)$ is an indicator of significantly selected neighbors for the p[th] word in the segment k, and thus $\sum_{p\sim\tilde{p}|k} w_{p\tilde{p}}\, I(z_{k\tilde{p}} = 1)$ provides the total weighted number of selected neighbors for the word p. Thus,

$$\log(R) = \log\pi\left(Z_{kp} = 0 \middle| all\ others\ except\ \beta_{kp}\right)$$

$$-\log\pi\left(Z_{kp} = 1 \middle| all\ others\ except\ \beta_{kp}\right)$$

$$= \left\{ -\frac{1}{2}\log\left|\sigma^2\mathbb{I}_{n_k}\right| - \frac{1}{2}\left(\underline{Y}_{[id_k]} - \underline{X}^T_{id_k[-p]}\underline{\beta}_{k[-p]}\right)^T \left(\sigma^2\mathbb{I}_{n_k}\right)^{-1}\left(\underline{Y}_{[id_k]} - \underline{X}^T_{id_k[-p]}\underline{\beta}_{k[-p]}\right)\right\}$$

$$-\left\{ \lambda \sum_{p\sim\tilde{p}|k} w_{p\tilde{p}}\, I(z_{k\tilde{p}} = 1) - \frac{1}{2}\log\left|\underline{X}_{id_k[p]}\underline{X}^T_{id_k[p]}\tau_p^2 + \sigma^2\mathbb{I}_{n_k}\right| \right.$$

$$\left. -\frac{1}{2}\left(\underline{Y}_{[id_k]} - \underline{X}^T_{id_k[-p]}\underline{\beta}_{k[-p]}\right)^T \left(\underline{X}_{id_k[p]}\underline{X}^T_{id_k[p]}\tau_p^2 + \sigma^2\mathbb{I}_{n_k}\right)^{-1}\left(\underline{Y}_{[id_k]} - \underline{X}^T_{id_k[-p]}\underline{\beta}_{k[-p]}\right)\right\}$$

$$= -\lambda \sum_{p\sim\tilde{p}|k} w_{p\tilde{p}}\, I(z_{k\tilde{p}} = 1) - \frac{1}{2}\left(\log\left|\sigma^2\mathbb{I}_{n_k}\right| - \log\left|\underline{X}_{id_k[p]}\underline{X}^T_{id_k[p]}\tau_p^2 + \sigma^2\mathbb{I}_{n_k}\right|\right)$$

$$-\frac{1}{2}\left(\underline{Y}_{[id_k]} - \underline{X}^T_{id_k[-p]}\underline{\beta}_{k[-p]}\right)^T \left[\left(\sigma^2\mathbb{I}_{n_k}\right)^{-1} - \left(\underline{X}_{id_k[p]}\underline{X}^T_{id_k[p]}\tau_p^2 + \sigma^2\mathbb{I}_{n_k}\right)^{-1}\right]\left(\underline{Y}_{[id_k]} - \underline{X}^T_{id_k[-p]}\underline{\beta}_{k[-p]}\right).$$

An application of the Sherman-Morrison-Woodbury formula yields:

$$\left(\sigma^2\mathbb{I}_{n_k}\right)^{-1} - \left(\underline{X}_{id_k[p]}\underline{X}^T_{id_k[p]}\tau_p^2 + \sigma^2\mathbb{I}_{n_k}\right)^{-1} = \frac{1}{(\sigma^2)^2}\underline{X}_{id_k[p]}\left[\frac{1}{\tau_p^2} + \frac{1}{\sigma^2}\underline{X}^T_{id_k[p]}\underline{X}_{id_k[p]}\right]^{-1}\underline{X}^T_{id_k[p]}.$$

Also, $\left|\underline{X}_{id_k[p]}\underline{X}^T_{id_k[p]}\tau_p^2 + \sigma^2\mathbb{I}_{n_k}\right| = \left|\sigma^2\mathbb{I}_{n_k}\right|\left(1 + \frac{\tau_p^2}{\sigma^2}\underline{X}^T_{id_k[p]}\underline{X}_{id_k[p]}\right)$.

Hence:

$$\log(R) = -\lambda \sum_{p\sim\tilde{p}|k} w_{p\tilde{p}}\, I\left(z_{k\tilde{p}} = 1\right) + \frac{1}{2}\log\left(1 + \frac{\tau_p^2}{\sigma^2}\underline{X}_{id_k[p]}^T \underline{X}_{id_k[p]}\right) -$$

$$\frac{1}{2(\sigma^2)^2}\left(\underline{Y}_{[id_k]} - \underline{X}_{id_k[-p]}^T \underline{\beta}_{k[-p]}\right)^T \underline{X}_{id_k[p]} \left[\frac{1}{\tau_p^2} + \frac{1}{\sigma^2}\underline{X}_{id_k[p]}^T \underline{X}_{id_k[p]}\right]^{-1} \underline{X}_{id_k[p]}^T \left(\underline{Y}_{[id_k]} - \underline{X}_{id_k[-p]}^T \underline{\beta}_{k[-p]}\right).$$

In this equation for $\log(R)$ above, if the counted number of selected neighbors, $\sum_{p\sim\tilde{p}|k} w_{p\tilde{p}} I\left(z_{k\tilde{p}} = 1\right)$, is positively increased, the term of $-\lambda \sum_{p\sim\tilde{p}|k} w_{p\tilde{p}} I\left(z_{k\tilde{p}} = 1\right)$ will be decreased as a negative number and $\log(R)$ will be decreased by the impact of the term. Then, the selection probability of p[th] word in the k[th] segment will be increased as shown from the equation, $\pi\left(Z_{kp} = 1 \middle| \sum_{p\sim\tilde{p}} w_{p\tilde{p}} I\left(z_{\tilde{p}} = 1\right),\ \beta_{k[-p]}, \tau_p^2, \sigma^2\right) = \frac{1}{1+R}$. With the mechanism, as the number of selected neighbors of p[th] word is increased, the probability of selecting p[th] word is increased.

(5) Full conditional distribution of τ_p^2:

$$\pi\left(\tau_{\mathrm{p}}^2 \middle| all\ others\right) = \pi\left(\underline{\beta} \middle| Z_{kp}, \tau_{\mathrm{p}}^2\right)\pi\left(\tau_{\mathrm{p}}^2\right)$$

$$\propto \left[\prod_{k=1:K;Z_{kp}=1} \frac{1}{\sqrt{\tau_{\mathrm{p}}^2}} e^{-\frac{\beta_{kp}^2}{2\tau_{\mathrm{p}}^2}}\right]\left(\tau_{\mathrm{p}}^2\right)^{-s_{p1}-1} e^{-\frac{s_{p2}}{\tau_{\mathrm{p}}^2}}$$

$$\propto \left(\frac{1}{\tau_{\mathrm{p}}^2}\right)^{\frac{\sum_{k=1}^K \mathbf{1}_{\{Z_{kp}=1\}}}{2} + s_{p1} + 1} e^{-\frac{\sum_{k=1}^K \beta_{kp}^2 \mathbf{1}_{\{Z_{kp}=1\}} + 2s_{p2}}{2\tau_{\mathrm{p}}^2}}$$

$$\pi\left(\tau_p^2 \middle| all\ others\right) = InvGamma\left(s_{p1} + \frac{\sum_{k=1}^K \mathbf{1}_{\{Z_{kp}=1\}}}{2}, s_{p2} + \frac{\sum_{k=1}^K \beta_{kp}^2 \mathbf{1}_{\{Z_{kp}=1\}}}{2}\right).$$

(6) Full conditional distribution of σ^2:

$$\pi(\sigma^2 | all\ others) = f\left(\underline{Y} \middle| \underline{\beta}, \underline{\mathrm{H}}, \sigma^2\right)\pi(\sigma^2)$$

$$\propto \left[\prod_{i=1}^{n} \sigma^{-1} e^{-\frac{\left(Y_i - \underline{X}_i^T \underline{\beta}_{H_i}\right)^2}{2\sigma^2}}\right] (\sigma^2)^{-\gamma_1 - 1} e^{\frac{-\gamma_2}{\sigma^2}}$$

Thus,

$$\pi(\sigma^2 | all\ others) = InvGamma\left(r_1 + \frac{n}{2}, r_2 + \frac{\sum_{i=1}^{n}\left(Y_i - \underline{X}_i^T \underline{\beta}_{H_i}\right)^2}{2}\right).$$

An alternative approach for co-estimating λ

As described at (Smith and Farhmeir 2007), λ, a hyper-parameter for strength of dependence energy, can be empirically determined a priori using model selection criteria (Higdon 1998). But as an alternative approach, co-estimation of λ_p can be considered as follows. First, note that $\pi(\lambda_p, Z_p) = \pi(Z_p|\lambda_p)\pi(\lambda_p)$. Assuming $\pi(\lambda_p) \propto I(0 < \lambda_p < \lambda_{max})$, where λ_{max} is a prespecified maximum ($\lambda_{max} = 1$ in this paper). Then,

$$\pi(\lambda_p | Z_p, \underline{Y}) \propto \pi(Z_p|\lambda_p) I(0 < \lambda_p < \lambda_{max})$$

(A-1)
$$\propto \frac{1}{Q_p(\lambda_p)} \exp\left(\lambda_p \textstyle\sum_{p\sim\tilde{p}} w_{p\tilde{p}} z_p z_{\tilde{p}}\right) I\left(0 < \lambda_p < \lambda_{max}\right).$$

$$Q_p(\lambda_p)^{-1} = \sum_{Z_p} \exp\left(\lambda_p U(Z_p)\right),$$

here $U(Z_p) = \sum_{p\sim\tilde{p}} w_{p\tilde{p}} z_p z_{\tilde{p}}$. Because the normalizing constant is difficult to compute analytically, Q_p is empirically approximated (see Smith and Farhmeir 2007, Gelman and Meng 1998). If $\lambda_p > 0$, $Q_p(\lambda_p) \propto \exp(-c(\lambda_p))$. Here, the $c_p(\lambda_p) = \int_0^{\lambda} E\left(U(Z_p)\middle|\lambda'\right) d\lambda'$ and the integral term of $c_p(\lambda_p)$ can be approximated using the following equation (A-2). For each λ' in a discrete grid over the domain [0, λ_{max}], calculate

(A-2)
$$E\left(U(\underline{Z})\middle|\lambda_p = \lambda_p'\right) \approx \frac{1}{J}\sum_{j=1}^{J} U\left(\underline{Z}^{(j)}\right);\ \underline{Z}^{(j)} \sim \pi\left(\underline{z}|\lambda_p = \lambda_p'\right).$$

The iterates $\underline{Z}^{(1)}, \ldots, \underline{Z}^{(J)}$ are obtained through Gibbs Sampling from the Ising prior conditioned on $\lambda_p = \lambda_p'$ (using *IsingSampler* function in R package *IsingSampler*). Second, using these values, the $E\left(U\left(\underline{Z}\right)\middle|\lambda_p\right)$ can be approximated over [0, λ_{max}] using a numerical interpolation. Similarly, for any value of $\lambda_p \in [0, \lambda_{max}]$, an approximate value of $c_p(\lambda_p)$ can be found and normalizing constant (Q_p) can be approximated with $Q_p\left(\lambda_p\right) \propto \exp(-c_p(\lambda_p))$. Now, with the numerically approximated Q_p, we can sample λ_p from the Equation (A-1) by the random-walk Metropolis-Hastings step. In particular, a proposal λ_p^{new} is derived from the λ_p^{old} by adding a small normal error (e.g., N(0, $(0.2)^2$) following Smith and Farhmeir 2007). The proposal can be accepted over the old one with probability:

$$\min\left(1, \exp\left\{c\left(\lambda_p^{old}\right) - c\left(\lambda_p^{new}\right) + \left(\lambda_p^{new} - \lambda_p^{old}\right)U\left(Z_p\right)\right\}\right).$$

Reference

Gelman, Andrew and Xiao-Li Meng (1998), "Simulating Normalizing Constants: From Importance Sampling to Bridge Sampling to Path Sampling," *Statistical Science*, 13, 163-185

Higdon, Dave (1998), "Auxiliary Variable Selection Methods for Markov Chain Monte Carlo with Applications," *Journal of American Statistical Association*, 93, 585–593.

Smith, Michael, and Ludwig Fahrmeir (2007), "Spatial Bayesian Variable Selection with Application to Functional Magnetic Resonance Imaging," *Journal of American Statistical Association*, 102, 417–431.

Web Appendix C: Steps of the Proposed MCMC Estimation Algorithm

Step 1. Conduct preprocessing to clean the unstructured texts and build a word-document frequency matrix (see the Web Appendix A).

Step 2. Initialize parameters of σ^2, $\underline{d}$, H_i, $\underline{\beta}_k$, $\underline{Z}_k$, and $\underline{\tau}^2$, and prespecify a value for λ (as an alternative approach, λ can be co-estimated as described in Web Appendix B).

Step 3. Generate H_i with $\pi(H_i|all\ others) = Multinoulli(d_1^*, \dots, d_K^*)$,

$$\text{where } d_k^* = \frac{\exp\left(-\frac{(Y_i - \underline{X}_i^T \underline{\tilde{\beta}}_k)^2}{2\alpha^2}\right) d_k}{\sum_{l=1}^{K} \exp\left(-\frac{(Y_i - \underline{X}_i^T \underline{\tilde{\beta}}_l)^2}{2\alpha^2}\right) d_l}.$$

Step 4. Update $\underline{d}$ from $Dirichlet(a_1 + n_1, \dots, a_K + n_K)$, where n_k is the number of members in the kth segment.

Step 5. Jointly sample $\underline{Z}_k$ and $\underline{\beta}_k$ for $k = 1, \dots, K$:

a. Update $\sum_{p \sim \tilde{p}|k} w_{p\tilde{p}} I(z_{k\tilde{p}} = 1)$, sequentially for $p = 1, \dots, P$. For doing this, compute the pairwise distances between each pth word and all other words in the kth segment with the word occurrence matrix (i.e., if a word occurs in the ith document, the corresponding cell of the matrix is 1, otherwise 0). With the computed distances, decide a set of neighbor words with a quantile threshold (e.g., words lower than a median distance used in the current application studies). Then, compute the weighted number of the selected neighbors of word p, that is $\sum_{p \sim \tilde{p}|k} w_{p\tilde{p}} I(z_{k\tilde{p}} = 1)$.

b. Sample Z_{kp} sequentially for $p = 1, \dots, P$ with:

$$\pi(Z_{kp}|all\ others\ except\ \beta_{kp}) = Bernoulli\left(\frac{1}{1+R}\right),$$

where $\log(R) = -\lambda \sum_{p \sim \tilde{p}|k} w_{p\tilde{p}} I(z_{k\tilde{p}} = 1) + \frac{1}{2}\log\left(1 + \frac{\tau_p^2}{\sigma^2} \underline{X}_{id_k[p]}^T \underline{X}_{id_k[p]}\right) -$

$$\frac{1}{2(\sigma^2)^2}\left(\underline{Y}_{[id_k]}-\underline{X}^T_{id_k[-p]}\underline{\beta}_{k[-p]}\right)^T\underline{X}_{id_k[p]}\left[\frac{1}{\tau_p^2}+\frac{1}{\sigma^2}\underline{X}^T_{id_k[p]}\underline{X}_{id_k[p]}\right]^{-1}\underline{X}^T_{id_k[p]}\left(\underline{Y}_{[id_k]}-\underline{X}^T_{id_k[-p]}\underline{\beta}_{k[-p]}\right),$$

where $I(z_{k\tilde{p}}=1)$ is an indicator of significantly selected neighbors for the p[th] word in the segment *k*, and thus $\sum_{p\sim\tilde{p}|k}w_{p\tilde{p}}I(z_{k\tilde{p}}=1)$ provides the weighted number of significantly selected neighbors for the p[th] word. In this equation for log (R), if a value of $\sum_{p\sim\tilde{p}|k}I(z_{k\tilde{p}}=1)$ is increased to a large positive numbers, a value of log (R) will be decreased with $-\lambda\sum_{p\sim\tilde{p}|k}w_{p\tilde{p}}I(z_{k\tilde{p}}=1)$ and then selection probability of p[th] word will be increased (see Bernoulli $\left(\frac{1}{1+R}\right)$ above). In addition, id_k is a set of members in the segment *k* (i.e., $id_k=\{i:H_i=k\}$), $\underline{X}^T_{i[-p]}$ is $\underline{X}^T_i$ without the p[th] word, and $\underline{\beta}_{k[-p]}$ is $\underline{\beta}_k$without its p[th] word.

c. If $Z_{kp}=0$, then $\pi(\beta_{kp}=0|Z_{kp}=0\ and\ others)=1$.

If $Z_{kp}=1$ (i.e., when a p[th] word in the segment k is selected), then generate β_{kp} with

$$\pi(\beta_{kp}|Z_{kp}=1\ and\ others)=N\left(\frac{\frac{1}{\alpha^2}\sum_{i\in id_k}\left(Y_i-\underline{X}^T_{i[-p]}\underline{\beta}_{k[-p]}\right)X_{ip}}{\frac{1}{\tau_p^2}+\frac{1}{\alpha^2}\sum_{i\in id_k}X_{ip}^2},1/\left(\frac{1}{\tau_p^2}+\frac{1}{\alpha^2}\sum_{i\in id_k}X_{ip}^2\right)\right).$$

Step 6. Generate τ_p^2 with

$$\pi(\tau_p^2|all\ others)=InvGamma\left(s_{p1}+\frac{\sum_{k=1}^K\mathbf{1}_{\{Z_{kp}=1\}}}{2},s_{p2}+\frac{\sum_{k=1}^K\beta_{kp}^2\mathbf{1}_{\{Z_{kp}=1\}}}{2}\right).$$

Step 7. Generate σ^2 with:

$$\pi(\sigma^2|all\ others)=InvGamma\left(r_1+\frac{n}{2},r_2+\frac{\sum_{i=1}^n\left(Y_i-\underline{X}^T_i\underline{\beta}_{H_i}\right)^2}{2}\right).$$

Step 8. Apply the relabeling algorithm for parameters $\underline{Z_k}, \underline{H}, \underline{d}$ through the 1,…, K segments following $\beta_{1p} < \beta_{2p} < \cdots < \beta_{Kp}$ and save the posterior samples per MCMC iteration. The posterior results (e.g., $\underline{\beta}$, $\underline{Z}$ and $\underline{H}$) are saved through iterations. Especially, the list of selected neighbor words of each word per each segment are saved per iteration.

Step 9. Determine the number of segments (K) and λ with model selection criteria (e.g., log marginal likelihood). Using the saved selected neighbors, build a hierarchical clustering heatmap to identify segment-specific groups of frequently co-occurring words (i.e., topics).

Web Appendix D: Monte Carlo Simulation Study

Simulation Study Design

Before applying the newly developed model to real-world application datasets, we confirmed its validity by conducting a Monte Carlo simulation study. This study assessed the model's capability to recover the true segment-level regression coefficients and identify predefined sets of group variables (i.e., true topics), given a set of simulation datasets, where data structure, true parameters, and true group structure of words were known (e.g., DeSarbo and Cron, 1988; Jedidi and DeSarbo, 1991; Kim et al., 2018). We generated a collection of synthetic datasets by manipulating independent factors reflecting different data structures and parameters (e.g., regression coefficients, predefined group variables). We employed a $3 \times 2 \times 2 \times 3 = 36$ full factorial design for a total of 36 simulation runs to study the main effects of four experimentally manipulated factors (with two or three levels per factor): the number of segments (i.e., K = 2, K = 3, or K = 4; Factor 1), the number of reviews/documents (i.e., sample size by N = 5000 or N = 10000; Factor 2), the number of words (i.e., the number of predictors included in the regression, P = 200 or P = 400; Factor 3), and the number of words within each predefined, significant topic (i.e., D = 3, D = 6, or D=10; Factor 4). In this simulation, we simply assume one topic per segment, but the topic can include various numbers of words as described in the condition of Factor 4.

For each condition, we first stochastically generated a word-document frequency matrix (independent variables matrix; $\underline{X}$) across segments using the *sample* function in R. Particularly, we predefined a group of words (e.g., 3, 6, or 10 words included in a topic) as a set of topic-related words. In the frequency matrix ($\underline{X}$), the frequently co-occurring group words were sampled with higher frequency probability (e.g., 90%) to simulate high co-occurrence structure

and other non-topic words were sampled with relatively lower frequency probability (e.g., 40%). In that way, the predefined words within a topic were co-occurring more frequently than other words. Once the word-document frequency matrix was generated, we also generated the true set of regression coefficients that included sparse portion of active variables (i.e., about 10% to 20% coefficients significantly away from zero) per segment. Here the predefined topic words were part of the active variables. Then, using the $\underline{X}$ with group variable structure, the segment-level true coefficients and a randomly generated membership vector (H_i), we generated dependent variables ($\underline{Y}$) by using the model equation of $Y_i = \underline{X}_i^T \underline{\beta}_{H_i} + \varepsilon_i,\ i = 1, \ldots, n$. After that, by submitting the $\underline{X}$ and $\underline{Y}$ to the proposed model, we could assess whether the true coefficients and true set of topic words were successfully identified.

We used three different metrics to quantify model performance: (i) the mean absolute error (MAE) between the true coefficients and their estimates, (ii) the mean proportion of misclassified topic words (False-Missing %), and (iii) the mean proportion of falsely classified non-topic words (False-Inclusion %). For example, if one word out of ten true topic words is missing in the estimated set of topics, the model achieves 10% False-Missing. Likewise, if the estimated set of topic words does not include any non-topic word, the model achieves 0% False-Inclusion. For MCMC estimation, we performed 10,000 iterations and discarded the first 5,000 as burn-in; we checked trace plots to check convergence.

Sensitivity Analyses of Two Tuning Parameters

To understand influence of two tuning parameters (i.e., the quantile threshold value for deciding neighbors and the tuning parameter λ) on the model estimates, we conducted sensitivity analyses using simulation studies with various conditions.

The first tuning parameter is the quantile threshold value for deciding neighbors versus non-neighbors. Intuitively, with a small threshold value, analysts have a narrow set of strong neighbors, but with a large threshold value, they have a broad set of neighbors. We first tried two extreme cases (i.e., 0 or 1) for the value to understand its influence. When the quantile value was zero, no word was counted as a neighbor, which eliminates False-Inclusion (0%) at the cost of increased likelihood of False-Missing (80%). (See Table D-1). In contrast, when the quantile value was one, every selected word was decided as a neighbor, which increases False-Inclusion as 89% while preventing a False-Missing as 0%.

To determine a quantile threshold value, we suggest a median quantile value of 0.5 as a default value. We found that any quantile value between 0.35 and 0.6 constrained both the False-Missing and False-Inclusion percentages to reasonably small values (i.e., 0% for False-Missing and 4% to 28% for False-Inclusion; see Table D-1). As a managerial purpose, if marketers want to identify narrow and strong neighbors within topics, they can use a small quantile value for the threshold (e.g., 0.25 to 0.35). But, if marketers want to include broad and weak neighbors within topics, they can use a large quantile value for the threshold (e.g., 0.65 to 0.75).

--- Insert Table D-1 about here ---

The second parameter is λ to control the network effect by neighbors for a word selection (see Equation (2)). If we set λ as zero, the proposed model does not consider any effects by neighbors but simply conducts variable selection independently, which resembles Kim et al.'s approach (2012). But, as λ increases, the effect by neighbors strengthens for selecting a word. In this simulation study, when we used a small λ (e.g., 0.01), the true active variables were correctly selected. However, when we increased the λ value (e.g., 0.5), some extra variables were selected, and when we use a large λ value (e.g., 1 or higher), all variables were incorrectly

selected due to the strong network power among words. To explain this, when we use a very high λ, all variables are tightly and strongly connected as neighbors and so the strong network power between neighbors can make all variables to be selected all together. For deciding λ value, we empirically choose a best λ value by using log marginal likelihood criteria by Chib's method (1995) (See Table 5 in the paper).

In sum, the sensitivity analyses show how the quantile threshold value controls False-Missing (%) and False-Inclusion (%) and how λ affects a word selection by controlling the effect by its neighbors. Based on the sensitivity analyses with respect to the two tuning parameters, we set the quantile threshold value to be a median of 0.5 and λ to be 0.01.

Monte Carlo Simulation Study Results

Using three metrics of MAE, False-Missing (%) and False-Inclusion (%), we confirmed that the proposed model successfully recovered the true parameters (i.e., segment-level coefficients) as well as identified the true set of words in the predefined topics. As presented at the Table D-2, consistently across all 36 experimental conditions, the estimates of the proposed model show a good match with the true parameter values (MAE=0.0064). Moreover, the proposed model successfully identifies the true topic words, as is shown in a small chance of misclassification (e.g., 2.4% False-Missing and 26.8% False-Inclusion). Note that the small error for classification of topic words is expected because the word-document frequency matrix was generated probabilistically (e.g., with a higher 90% probability for topic words and with a lower 40% probability for non-topic words).

We also checked the robustness of the model's performance by running multiple linear regression analyses (akin to ANOVA) on the three dependent metrics after coding the four

independent factors (i.e., Factor 1 to Factor 4) as six dummy independent variables to explore their main effects. As presented in Table D-3, none of the independent factors were significant at a 5% significance level for the MAE dependent metrics, except when the number of segments increased from K=2 to K=4. Although the MAE increased by the K=4, the MAE for the condition of K=4 was still quite small at 0.011. Next we could not find any significant independent factors for False-Missing (%) across conditions, as the proposed model shows almost perfect performance in the measure. Last, none of the independent factors were significant for the False-Inclusion (%), except when the number of predictors was increased from P=200 to P=400. Unsurprisingly, as the number of predictors significantly increased from 200 to 400, False-Inclusion (%) increased from 24.9% to 28.6% (see Table D-2).

In sum, this Monte Carlo simulation study demonstrates that the proposed method successfully recovers the true segment-level regression coefficients and identifies the true set of predefined groups of words included in each topic. In addition, it also demonstrates a reasonably good robustness with respect to all four experimental factors and their two or three levels for three dependent measures except only one significant dummy for MAE and for False-Inclusion %.

--- Insert Tables D-2 and D-3 about here ---

References

Chib, Siddhartha (1995), "Marginal Likelihood from the Gibbs Output," *Journal of the American Statistical Association*, 90 (432), 1313-21.

DeSarbo, Wayne S., and William L. Cron (1988), "A Maximum Likelihood Methodology for Clusterwise Linear Regression," *Journal of Classification*, 5 (2), 249–282.

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Table D-1. Sensitivity analysis with a spectrum of quantile threshold values for deciding neighbors using a simulation study of K=3 condition.

Quantile threshold values	MAE	False-Inclusion (%)	False-Missing (%)
0	0.0072	0%	80%
0.1	0.0073	0%	67%
0.2	0.0072	0%	29%
0.3	0.0067	0%	9%
0.35	0.0071	4%	0%
0.4	0.0066	13%	0%
0.45	0.0069	22%	0%
0.5	0.0069	23%	0%
0.55	0.0068	24%	0%
0.6	0.0069	28%	0%
0.65	0.0068	36%	0%
0.7	0.0070	36%	0%
0.8	0.0067	58%	0%
0.9	0.0067	82%	0%
1	0.0078	89%	0%

Table D-2. Simulation study results across two or three levels of four experimental factors.

		Experimentally Manipulated Factors			
	Level	Factor 1 (Number of Segments)	Factor 2 (Sample Size)	Factor 3 (Number of Predictors)	Factor 4 (Number of Words per Topic)
MAE of segment-level regression coefficient recovery	1	0.0031	0.0077	0.006	0.008
	2	0.0047	0.005	0.0067	0.006
	3	0.0112	NA	NA	0.005
	Average	0.0064	0.0064	0.0064	0.0064
False-Missing (%)	1	0.012	0.031	0.01	0.038
	2	0.025	0.017	0.039	0.021
	3	0.034	NA	NA	0.013
	Average	0.024	0.024	0.024	0.024
False-Inclusion (%)	1	0.269	0.264	0.249	0.264
	2	0.277	0.272	0.286	0.273
	3	0.257	NA	NA	0.266
	Average	0.268	0.268	0.268	0.268

Note: We have “NA” for level 3 of Factor 2 and Factor 3, because Factors 2 and 3 do not have level 3 conditions.

Table D-3. Linear model result on three dependent measures with four experimental factors.

Independent Dummy Variables of Linear Model	p-value for the recovery of coefficients	p-value for the recovery of False-Missing (%)	p-value for the recovery of False-Inclusion (%)
Factor 1, dummy 1 (K=3)	p = 0.361	p = 0.687	p = 0.665
Factor 1, dummy 2 (K=4)	p < 0.05 *	p = 0.487	p = 0.477
Factor 2 (N=10,000)	p = 0.051	p = 0.592	p = 0.596
Factor 3 (P=400)	p = 0.591	p = 0.237	p < 0.05 *
Factor 4, dummy 1 (D=6)	p = 0.068	p = 0.433	p = 0.933
F4, dummy 2 (D=10)	p = 0.440	p = 0.799	p = 0.648

Note: For dummy variables coding with 3 levels, we choose the first level (e.g., K=2 and D=3) as a reference group; Factor 1 refers to number of segments, Factor 2 refers to sample size, Factor 3 refers to number of predictors, and Factor 4 refers to number of words per topic.

Web Appendix E: More Tables in Empirical Application Study of Restaurant Segmentation with Yelp Review Data

Table E-1. Aggregate regression result with all 190 words for Yelp review data.

Words	Coeffs	P-value	Words	Coeffs	P-value	Words	Coeffs	P-value
(Intercept)	3.52	0	bad	-0.9	0	want	0.02	0.68
burger	-0.03	0.16	drink	-0.08	0.04	come	0.12	0.01
food	-0.07	0	night	-0.09	0.11	two	0.05	0.43
good	0.12	0	call	-0.18	0	around	0.04	0.6
great	0.56	0	never	-0.52	0	lot	0.06	0.4
order	-0.08	0	bowl	0.02	0.64	pretti	-0.28	0
place	-0.02	0.33	even	-0.24	0	cold	-0.69	0
ask	-0.21	0	minut	-0.25	0	fish	-0.11	0
get	0.02	0.51	waitress	-0.23	0	right	0.09	0.19
just	-0.13	0	bbq	-0.06	0.17	seem	-0.52	0
chicken	-0.05	0.04	pork	-0.04	0.41	sinc	-0.04	0.56
like	-0.18	0	owner	0.19	0	walk	-0.12	0.11
time	-0.04	0.28	beer	-0.05	0.08	everi	0.43	0
red	0.06	0.38	select	-0.03	0.69	top	0.07	0.28
servic	-0.08	0.06	chees	-0.09	0.04	sauce	0.18	0.06
fresh	0.34	0	littl	0.17	0	tabl	-0.07	0.25
salad	-0.09	0	love	0.65	0	guy	-0.02	0.81
tri	0.3	0	wine	-0.02	0.56	now	0.05	0.52
sandwich	-0.02	0.33	server	-0.15	0.02	kid	-0.25	0
beef	-0.06	0.19	meat	0.02	0.68	told	-0.18	0.03
fri	-0.05	0.11	egg	-0.05	0.43	long	-0.12	0.16
hot	0.04	0.33	rice	-0.07	0.1	patio	-0.25	0
lunch	-0.01	0.9	shrimp	-0.03	0.52	new	0.05	0.3
roll	-0.02	0.58	cream	0.09	0.15	clean	0.21	0
sushi	-0.04	0.15	ice	-0.05	0.48	room	-0.01	0.9
alway	0.23	0	give	-0.4	0	husband	0.17	0.03
came	0.13	0.02	got	-0.11	0.02	sure	0.38	0
soup	-0.02	0.66	fast	0.01	0.95	think	-0.14	0.03
will	0.17	0	first	0.15	0.01	definit	1.07	0
line	-0.28	0	open	-0.22	0	park	0.03	0.76
thai	0	0.9	peopl	-0.15	0.02	see	-0.03	0.67
drive	-0.35	0	staff	0.08	0.14	special	-0.07	0.25

taco	0.04	0.1	friend	0.34	0	find	0.3	0
back	-0.06	0.14	make	0.17	0	tea	0	0.99
dish	0.12	0.01	sweet	-0.04	0.52	live	0	0.97
one	-0.22	0	cook	-0.11	0.09	music	-0.06	0.37
price	-0.13	0.01	eat	-0.16	0	area	0.05	0.46
realli	-0.06	0.16	home	0.12	0.09	side	0.11	0.08
tast	-0.26	0	review	-0.23	0	serv	-0.17	0.01
wait	0.03	0.51	take	0.22	0	grill	0.14	0.01
dinner	-0.04	0.5	year	0.32	0	much	-0.08	0.19
steak	0.02	0.57	happi	0.06	0.4	help	0.43	0
burrito	-0.01	0.7	hour	-0.2	0.01	made	0.05	0.38
pizza	-0.01	0.3	spici	0.19	0.01	bacon	0.12	0.05
breakfast	-0.08	0.03	coffe	0.04	0.21	dessert	0.11	0.15
fun	0.22	0	also	0.21	0	took	-0.15	0.06
mexican	0.02	0.65	flavor	0.07	0.15	tell	-0.02	0.86
said	-0.25	0	day	0.15	0.02	ever	-0.4	0
bar	-0.13	0	look	-0.12	0.01	still	-0.15	0.04
salsa	-0.17	0.01	star	-0.02	0.77	way	-0.1	0.14
best	1.05	0	work	0.13	0.04	bean	-0.09	0.21
close	-0.12	0.15	nice	0.04	0.33	enjoy	0.28	0
locat	-0.02	0.73	old	-0.3	0	better	-0.38	0
manag	-0.58	0	custom	-0.14	0.04	offer	-0.05	0.57
need	-0.14	0.05	know	-0.07	0.26	hous	0.17	0.03
water	-0.26	0	use	-0.34	0	dine	0.15	0.09
can	0.16	0	went	-0.29	0	big	0.06	0.44
wing	-0.06	0.07	bit	-0.01	0.9	perfect	0.59	0
restaur	-0.09	0.03	say	-0.17	0.01	potato	0.08	0.24
sauc	-0.19	0	small	-0.07	0.25	local	0.05	0.48
menu	0.05	0.25	meal	0.07	0.18	extra	-0.03	0.76
chip	-0.03	0.72	plate	0.06	0.38	famili	0.19	0.01
green	0.03	0.64	well	0.19	0	pay	-0.74	0
bread	-0.11	0	free	0.17	0.02			

Note, please see the p-value for significance.

Table E-2. K=3 result by latent class regression *without* variable selection (*flexmix* package in R) with all 190 words for Yelp review data.

	Segment 1		Segment 2		Segment 3	
Words	Coeffs	P-value	Coeffs	P-value	Coeffs	P-value
(Intercept)	3.51	0.00	3.94	0.00	3.14	0.00
burger	0.01	0.61	0.04	0.08	-0.03	0.56
food	-0.02	0.28	-0.04	0.09	-0.05	0.09
good	0.06	0.08	-0.10	0.01	0.61	0.00
great	0.59	0.00	0.42	0.00	0.55	0.00
order	0.01	0.76	-0.32	0.00	0.02	0.62
place	-0.07	0.01	0.27	0.00	-0.06	0.12
ask	-0.55	0.00	0.14	0.06	-0.49	0.00
get	0.16	0.00	-0.16	0.00	-0.07	0.38
just	-0.10	0.04	-0.28	0.00	-0.07	0.27
chicken	-0.14	0.00	0.08	0.01	0.02	0.61
like	-0.35	0.00	-0.13	0.00	0.03	0.56
time	-0.33	0.00	0.46	0.00	0.05	0.37
red	0.07	0.31	0.07	0.50	-0.49	0.00
servic	0.32	0.00	-0.43	0.00	-0.47	0.00
fresh	0.39	0.00	0.20	0.00	0.52	0.00
salad	-0.17	0.00	-0.11	0.00	0.03	0.51
tri	0.51	0.00	-0.21	0.00	0.46	0.00
sandwich	0.08	0.00	0.10	0.00	-0.14	0.00
beef	-0.27	0.00	-0.09	0.06	0.90	0.00
fri	-0.16	0.00	0.03	0.43	-0.15	0.03
hot	0.07	0.06	0.42	0.00	-0.30	0.00
lunch	-0.13	0.01	-0.02	0.61	0.20	0.00
roll	0.02	0.70	-0.05	0.45	0.07	0.50
sushi	-0.12	0.00	0.17	0.00	-0.23	0.00
alway	0.28	0.00	0.16	0.01	0.14	0.12
came	0.13	0.06	-0.28	0.00	0.49	0.00
soup	-0.03	0.57	-0.03	0.45	-0.20	0.01
will	-0.13	0.03	0.12	0.01	0.14	0.11
line	-0.08	0.46	-0.31	0.04	-0.15	0.21
thai	-0.04	0.17	0.01	0.84	0.10	0.10
drive	0.34	0.00	-1.27	0.00	-0.76	0.00
taco	-0.08	0.01	0.17	0.00	-0.09	0.08
back	0.28	0.00	-0.37	0.00	-0.02	0.80
dish	-0.01	0.83	0.09	0.08	0.02	0.72
one	0.00	0.94	-0.05	0.30	-1.02	0.00
price	-0.16	0.01	-0.16	0.00	0.04	0.65

realli	-0.14	0.00	0.33	0.00	-0.14	0.05
tast	-0.32	0.00	-0.51	0.00	-0.42	0.00
wait	-0.05	0.44	-0.07	0.25	0.26	0.02
dinner	0.18	0.01	0.15	0.05	-0.27	0.03
steak	0.10	0.06	-0.02	0.79	-0.11	0.09
burrito	-0.08	0.08	0.16	0.00	-0.12	0.09
pizza	-0.07	0.00	0.11	0.00	0.07	0.00
breakfast	-0.02	0.68	-0.11	0.00	-0.04	0.62
fun	0.03	0.78	-0.11	0.19	0.52	0.00
mexican	-0.04	0.52	-0.03	0.71	0.13	0.12
said	-0.10	0.21	-0.09	0.33	-0.52	0.00
bar	-0.11	0.00	-0.18	0.00	-0.10	0.19
salsa	-0.01	0.87	0.02	0.78	-0.67	0.00
best	0.90	0.00	0.61	0.00	1.98	0.00
close	-0.71	0.00	-0.02	0.89	0.75	0.00
locat	-0.33	0.00	0.22	0.02	0.24	0.03
manag	-1.22	0.00	0.05	0.66	0.47	0.00
need	-0.19	0.04	0.29	0.01	-0.74	0.00
water	-0.01	0.95	-1.15	0.00	0.16	0.32
can	-0.03	0.59	-0.07	0.20	1.00	0.00
wing	-0.06	0.18	-0.08	0.02	0.18	0.00
restaur	-0.32	0.00	0.09	0.09	0.10	0.16
sauc	-0.27	0.00	0.04	0.62	-0.97	0.00
menu	0.08	0.15	0.10	0.00	0.06	0.32
chip	-0.10	0.26	-0.23	0.03	0.60	0.00
green	0.14	0.03	0.01	0.89	-0.16	0.32
bread	-0.19	0.00	0.13	0.01	-0.25	0.00
bad	-1.21	0.00	-1.33	0.00	-0.28	0.01
drink	0.01	0.87	0.03	0.59	-0.44	0.00
night	-0.36	0.00	0.06	0.34	0.35	0.00
call	-0.11	0.16	-0.67	0.00	-0.32	0.00
never	-0.40	0.00	-0.65	0.00	-1.40	0.00
bowl	-0.12	0.03	-0.05	0.30	-0.08	0.32
even	-0.52	0.00	0.03	0.58	0.35	0.00
minut	-0.21	0.03	-0.22	0.00	-0.71	0.00
waitress	-0.10	0.18	-0.57	0.00	0.07	0.51
bbq	-0.03	0.52	-0.09	0.04	-0.04	0.61
pork	-0.24	0.00	0.47	0.00	-0.13	0.25
owner	0.06	0.27	0.22	0.00	0.38	0.00
beer	0.00	0.90	-0.04	0.28	-0.03	0.61
select	-0.37	0.00	0.30	0.00	-1.15	0.00

chees	-0.14	0.01	-0.15	0.01	-0.20	0.06
littl	0.01	0.91	0.33	0.00	-0.19	0.02
love	1.02	0.00	0.10	0.05	0.93	0.00
wine	0.08	0.03	0.02	0.67	-0.04	0.63
server	-0.27	0.00	0.00	0.96	0.16	0.16
meat	0.16	0.00	-0.42	0.00	0.14	0.00
egg	-0.12	0.14	0.11	0.09	0.09	0.36
rice	0.09	0.12	-0.18	0.00	-0.10	0.26
shrimp	-0.15	0.01	0.51	0.00	-0.19	0.06
cream	0.00	0.96	0.04	0.61	0.13	0.32
ice	0.05	0.56	-0.02	0.79	-0.17	0.20
give	-0.30	0.00	-0.03	0.76	-0.61	0.00
got	-0.21	0.00	-0.33	0.00	0.32	0.00
fast	0.04	0.74	-0.35	0.01	0.56	0.00
first	0.09	0.25	0.14	0.10	0.48	0.00
open	-0.23	0.00	-0.11	0.30	-0.33	0.00
peopl	0.08	0.36	-0.14	0.12	-0.62	0.00
staff	0.07	0.36	0.09	0.21	-0.35	0.00
friend	0.06	0.30	0.58	0.00	0.69	0.00
make	0.09	0.12	0.28	0.00	0.11	0.19
sweet	-0.03	0.72	-0.19	0.03	-0.29	0.03
cook	0.19	0.02	-0.14	0.09	-0.46	0.00
eat	-0.11	0.09	-0.17	0.01	-0.40	0.00
home	-0.06	0.47	-0.02	0.85	1.12	0.00
review	-0.04	0.67	0.32	0.00	-0.85	0.00
take	0.20	0.01	0.31	0.00	-1.16	0.00
year	0.23	0.05	-0.29	0.02	0.92	0.00
happi	0.35	0.00	-0.22	0.03	0.01	0.90
hour	-0.54	0.00	-0.25	0.01	0.27	0.02
spici	0.24	0.01	0.04	0.64	-0.10	0.49
coffe	0.03	0.50	-0.04	0.41	-0.03	0.53
also	0.38	0.00	-0.02	0.82	-0.07	0.43
flavor	0.13	0.05	0.23	0.00	-0.14	0.05
day	0.48	0.00	-0.09	0.29	0.18	0.13
look	0.29	0.00	-0.63	0.00	-0.48	0.00
star	0.14	0.14	-0.13	0.08	-0.20	0.21
work	0.08	0.29	-0.17	0.05	0.42	0.00
nice	0.06	0.29	0.01	0.83	-0.08	0.28
old	-0.14	0.06	-0.50	0.00	-0.76	0.00
custom	-0.15	0.05	-0.34	0.00	-0.28	0.03
know	0.08	0.30	0.26	0.00	0.07	0.49

use	-0.41	0.00	-0.46	0.00	-0.19	0.05
went	-0.18	0.02	-0.40	0.00	-0.60	0.00
bit	0.02	0.81	-0.17	0.10	-0.16	0.31
say	-0.03	0.63	-0.55	0.00	-0.56	0.00
small	-0.11	0.17	-0.06	0.37	-0.50	0.00
meal	-0.11	0.05	0.22	0.00	-0.16	0.14
plate	0.29	0.00	0.01	0.88	0.19	0.06
well	0.42	0.00	0.15	0.02	-0.39	0.00
free	0.20	0.02	0.05	0.69	0.80	0.00
want	0.18	0.01	-0.28	0.00	0.29	0.00
come	0.12	0.05	0.24	0.00	-0.03	0.72
two	-0.04	0.61	-0.15	0.06	0.16	0.22
around	0.42	0.00	-0.27	0.01	-0.15	0.43
lot	-0.42	0.00	0.59	0.00	0.66	0.00
pretti	-0.31	0.00	-0.01	0.90	-1.05	0.00
cold	-1.06	0.00	-0.80	0.00	-0.02	0.88
fish	0.01	0.90	-0.28	0.00	-0.07	0.25
right	0.61	0.00	0.00	0.97	-0.18	0.18
seem	-0.73	0.00	-0.43	0.00	0.50	0.00
sinc	-0.22	0.02	0.19	0.07	-0.61	0.00
walk	-0.37	0.00	-0.15	0.09	0.34	0.03
everi	-0.16	0.13	0.42	0.00	0.93	0.00
top	0.32	0.00	0.03	0.76	-0.07	0.64
sauce	0.80	0.00	-0.80	0.00	-0.87	0.00
tabl	-0.11	0.15	-0.30	0.00	-0.17	0.20
guy	-0.10	0.36	-0.85	0.00	0.85	0.00
now	0.56	0.00	-0.61	0.00	-0.01	0.97
kid	-0.16	0.05	-0.60	0.00	0.11	0.24
told	-0.46	0.00	-0.10	0.33	0.16	0.27
long	-0.18	0.10	0.49	0.00	-1.03	0.00
patio	-0.20	0.02	-0.11	0.18	-0.39	0.00
new	0.22	0.00	-0.18	0.01	0.10	0.27
clean	0.18	0.07	0.29	0.00	0.64	0.00
room	0.10	0.34	0.06	0.23	-0.48	0.00
husband	-0.31	0.00	0.60	0.00	0.75	0.00
sure	0.07	0.50	1.02	0.00	0.04	0.81
think	-0.56	0.00	0.53	0.00	-0.31	0.01
definit	1.14	0.00	0.85	0.00	1.92	0.00
park	-0.10	0.35	-0.02	0.88	0.83	0.00
see	-0.16	0.08	-0.01	0.89	0.34	0.01
special	0.08	0.37	0.01	0.92	0.09	0.51

find	0.25	0.00	-0.18	0.02	1.13	0.00
tea	-0.07	0.08	0.11	0.00	0.31	0.01
live	0.37	0.00	-0.02	0.80	-0.63	0.00
music	-0.31	0.00	-0.07	0.37	0.48	0.00
area	-0.02	0.77	0.36	0.00	-0.28	0.01
side	-0.02	0.76	0.21	0.01	-0.23	0.06
serv	-0.36	0.00	0.32	0.00	-0.39	0.00
grill	0.11	0.10	0.05	0.50	-0.30	0.00
much	0.33	0.00	-0.10	0.24	-0.31	0.01
help	0.66	0.00	0.71	0.00	-0.39	0.02
made	-0.18	0.02	-0.03	0.70	1.39	0.00
bacon	-0.05	0.40	0.07	0.18	-0.37	0.02
dessert	0.18	0.06	-0.12	0.28	0.76	0.00
took	-0.10	0.35	0.14	0.16	-0.48	0.00
tell	-0.07	0.59	0.64	0.00	-0.11	0.56
ever	-0.17	0.07	-0.15	0.16	-0.72	0.00
still	-0.06	0.52	-0.23	0.01	0.65	0.00
way	-0.25	0.01	-0.36	0.00	-0.25	0.01
bean	-0.22	0.03	-0.11	0.20	1.00	0.00
enjoy	0.40	0.00	-0.01	0.92	-0.05	0.63
better	-0.38	0.00	-0.33	0.00	-0.35	0.00
offer	0.28	0.01	-0.46	0.00	0.93	0.00
hous	0.30	0.01	-0.10	0.31	0.10	0.45
dine	0.10	0.39	-0.14	0.23	1.69	0.00
big	0.36	0.00	-0.05	0.65	-0.11	0.41
perfect	0.36	0.00	1.20	0.00	0.29	0.02
potato	0.15	0.03	0.27	0.01	-0.48	0.00
local	-0.26	0.01	0.02	0.83	0.63	0.00
extra	0.17	0.25	0.15	0.21	-0.49	0.00
famili	0.24	0.00	0.76	0.00	-1.11	0.00
pay	-0.66	0.00	-1.19	0.00	-0.79	0.00
Mixture	**47%**		**40%**		**14%**	

Note, please see the p-value for significance.

Table E-3. K=3 result by latent class regression *with* variable selection (Kim et al. 2012) with Yelp review data. A total of 80 words (average 38 words per segment; 20% of all 190 words were selected across three derived segments out of 190 words.

Segment 1 (34%)		Segment 2 (30%)		Segment 3 (36%)	
Words	Coeffs	Words	Coeffs	Words	Coeffs
good	0.21	good	0.286	good	-0.144
great	0.68	great	0.457	great	0.55
fresh	0.518	order	-0.18	order	-0.21
tri	0.438	like	-0.577	get	-0.26
fri	-0.206	fresh	0.447	just	-0.189
alway	0.507	will	0.536	servic	-0.385
one	-0.456	drive	-1.258	tri	0.353
tast	-0.485	back	-0.521	lunch	-0.185
best	1.275	price	-0.326	tast	-0.603
bad	-0.665	best	0.957	said	-0.524
never	-0.741	locat	-0.442	bar	-0.272
even	-0.553	manag	-0.998	best	0.926
owner	0.266	can	0.521	locat	0.431
littl	0.464	restaur	-0.289	need	-0.447
love	1.468	bad	-0.903	sauc	-0.337
give	-0.542	night	-0.262	bad	-1.363
first	-0.491	minut	-0.919	call	-0.699
peopl	-0.282	waitress	-0.517	owner	0.352
friend	0.318	chees	-0.286	love	0.314
make	0.454	love	0.426	open	-0.394
eat	-0.351	give	-0.542	also	0.377
home	0.491	first	1.065	day	0.347
review	-0.41	friend	0.405	old	-0.432
look	-0.306	flavor	0.373	use	-0.58
custom	-0.71	use	-0.387	went	-0.494
went	-0.542	say	-0.504	come	0.302
well	0.898	pretti	-0.774	cold	-1.436
cold	-0.909	everi	0.777	right	0.426
top	0.745	clean	0.597	seem	-1.39
now	0.504	definit	1.258	kid	-0.56
patio	-0.532	area	0.429	long	-0.579
clean	0.414	serv	-0.382	area	-0.382
definit	2.084	help	0.532	better	-0.444
ever	-0.807	way	0.58	perfect	0.939
way	-0.731	dine	0.613		
enjoy	0.696	perfect	0.717		
better	-0.642	pay	-1.746		
extra	-0.88				

Table E-4. The regression coefficients and significant group words (with a same color) of three segments selected by the proposed model with Yelp review data. A total of 96 words (average 45 words per segment; 23.5% of all 190 words) were selected across three derived segments out of 190 words.

Segment 1 (31%)		**Segment 2 (32%)**		**Segment 3 (37%)**	
Words	Coeffs	Words	Coeffs	Words	Coeffs
Intercept	2.691	Intercept	3.528	Intercept	4.167
good	0.334	**great**	**0.59**	great	0.446
great	0.671	**tri**	**0.412**	place	-0.13
place	0.224	**tast**	**-0.657**	**ask**	**-0.497**
ask	**-0.378**	best	1.218	like	-0.204
like	-0.325	bad	-1.101	**servic**	**-0.294**
servic	**0.328**	never	-0.467	will	0.338
tri	0.357	minut	-0.384	**best**	**0.786**
alway	**0.633**	**love**	**0.806**	**bad**	**-1.379**
will	0.454	**friend**	**0.324**	**never**	**-0.637**
drive	-0.586	**cook**	**0.396**	love	0.516
tast	**-0.358**	**eat**	**-0.235**	friend	0.556
fun	**0.853**	cold	-1.305	pretti	-0.691
bar	**-0.312**	seem	-0.433	**walk**	**-0.811**
best	**1.252**	walk	-0.668	**kid**	**-0.531**
close	**-0.487**	definit	1.005	definit	0.854
water	-0.53	**park**	**0.547**	**realli**	**-0.455**
bad	-0.554	**perfect**	**0.757**	**manag**	**-0.462**
night	**-0.436**	**fresh**	**0.713**	**can**	**0.532**
never	**-0.379**	salad	-0.153	custom	0.343
even	**-0.643**	back	-0.235	went	-0.436
minut	**-0.604**	one	-0.269	better	-0.792
chees	-0.286	realli	0.441	order	-0.293
love	**0.62**	said	-0.539	**chicken**	**-0.168**
give	-0.688	**locat**	**0.353**	**came**	**0.338**
got	**-0.551**	manag	-0.431	**wait**	**0.348**
friend	**0.318**	can	-0.302	**salsa**	**-0.249**
cook	-0.627	restaur	-0.225	**sauc**	**-0.545**
eat	**-0.291**	owner	0.343	**drink**	**-0.239**
also	**0.647**	littl	0.41	**call**	**-0.486**
look	-0.475	**open**	**-0.528**	**waitress**	**-0.447**
nice	**0.273**	make	0.393	**server**	**-0.589**
old	-0.405	star	-0.655	peopl	-0.342
meal	**0.347**	custom	-0.884	work	0.312
pretti	-0.382	use	-0.752	want	0.325

cold	-0.568	went	-0.431	right	0.412
seem	-0.836	**well**	**0.976**	everi	0.433
walk	0.843	**fish**	**-0.298**	**sauce**	**0.526**
kid	**-0.518**	top	0.612	special	-0.38
clean	0.828	**patio**	**-0.394**	find	0.553
sure	0.886	think	-0.459	**dine**	**0.364**
definit	1.484	**serv**	**-0.502**		
park	**-0.674**	much	-0.541		
see	-0.562	ever	-1.234		
live	**0.485**	way	-0.606		
help	**0.938**	**enjoy**	**0.474**		
dessert	**0.541**	better	-0.517		
perfect	1.2	famili	0.455		

Note: all shaded cells refer to selected words based on group variable selection; cells pertaining to a particular topic are shaded with orange, green, blue, yellow, or pink color; gray-shaded cells are selected as individual words.

Table E-5. Model comparison with Yelp review data between the proposed model and benchmark models

Learning Type	Unsupervised				Supervised				
Models	Topic Model Regression	Hierarchical Topic Model (HDP) Regression	Topic Model Regression *with* Interactions	Topics Model Latent Class regression *with* Interactions	Supervised Topic Model (McAuliffe and Blei 2010)	Aggregate Regression (OLS)	Latent Class Regression *without* Variable Selection	Latent Class Regression *with* Word Selection (Kim et al., 2012)	**The Proposed Model with Topic (grouped words) Selection**
Error for Prediction (MAE)	0.449	0.400	0.440	0.307	0.333	0.328	0.205	0.190	**0.188**
Recommendation Hit Rate (RHR)	82.6%	83.4%	82.6%	85.2%	84.5%	84.7%	88.1%	88.8%	**88.9%**
Coherence Score for interpretability (with UMass)	-0.389	-0.368	-	-	-0.404	-0.509	-0.524	-0.519	**-0.257**
Focus on Prediction ability	No	No	No	No	Yes	Yes	Yes	Yes	**Yes**
Ease of Interpretation through Identified Topics	Yes	Yes	Yes	Yes	Yes	No	No	No	**Yes**
Focus on Heterogeneity	No	No	No	Yes	No	No	Yes	Yes	**Yes**

Note, for MAE, lower value is better; for RHR higher rate value is better; for Coherence Score, a value to closer to zero means high coherence (i.e., better interpretability); coherence score is not available for the models with interaction terms.

Web Appendix F: Customer Segmentation Study for A Restaurant with the Largest Number of Reviews in the Yelp Dataset

Data Description

In this study, we demonstrate the proposed model's ability for customer-level segmentation, which has been a stream of segmentation studies in marketing (Danaher 1998, Kim et al. 2012). To conduct a segmentation analysis involving customers (reviewers), we identified a restaurant with the highest number of reviews in our Yelp review dataset - Pizzeria Bianco, which had a total of 2,036 customer reviews. The dependent variable was the star rating given by each customer. The overall mean of the ratings was 3.90. Among a total of 2,036 customers, 43.2% awarded five stars, 26.5% awarded four stars, 14.2% awarded three stars, 9.4% awarded two stars and 6.6% awarded one star. For independent variables, we used a word frequency matrix of 233 words, after excluding very sparse words that appeared in less than 5% of the reviews. Yelp did not provide any demographic or personal information (e.g., age, gender, or address) about the reviewers. However, certain review-related activities on Yelp were disclosed, such as the total review count, average review ratings, and the number of acknowledgements (e.g., usefulness) received from others.

Using the log marginal likelihood criteria suggested by Chib (1995), we selected a three-segment model with a 0.01 as a λ tuning parameter value as the best solution, as shown in Table F-1. To elucidate the co-occurring group structure among the selected words, we employed a heatmap, which is a popular method in biology/bioinformatics and many other fields for visualizing and interpreting underlying groups among elements (e.g., Haarman et al. 2015; Figure F-1).

--- Insert Table F-1 and Figure F-1 about here ---

Customer Segmentation Results

Table F-2 presents the identified topics and groups of neighbor words for each topic. As evidenced by the estimated coefficients in Table F-2, we observed a reasonably strong face validity such that the counts of positive words in a review such as “love,” “enjoy,” “amazing,” and “perfect” were positively correlated with a higher star rating (i.e., positive coefficients for these words). The negative coefficient associated with the term "better" might seem somewhat puzzling. However, we find this sign to be reasonable, as the word "better" is commonly used to signal an aspect that requires improvement, expressing a sense of dissatisfaction. We observed that neutral words like “bread,” “staff,” or “wiseguy” (a pizza name) can display either a positive or negative correlation depending on the review context. For example, the word “bread” in Segment 1 often appeared in lower-rated reviews (1- or 2- stars) to express dissatisfaction with bread quality, resulting in a negative coefficient of -0.3. In contrast, the word “wiseguy” (a pizza name) in Segment 3 was frequently used in higher-rated reviews (5- or 4- stars) to positively describe pizza quality, leading to a positive coefficient of 0.15. Therefore, through careful interpretation of the word-level coefficients provided by the proposed model, analysts can gain a more nuanced understanding of customer voices, thereby offering valuable insights for marketers. Table F-3 provides the regression coefficients for all selected words across the three segments.

--- Insert Tables F-2 and F-3 about here ---

Table F-4 briefly describes the noticeable characteristics of customer reviews for each segment and the profiles of reviewers belonging to it. First, Segment 1, which included 306 customers (15% of the total sample), had the lowest mean rating of 2.05 among all three

segments (i.e., 3.39 for Segment 2 and 4.71 for Segment 3; with $p < 0.01$). Thus, we named Segment 1 "Unhappy customer segment." The reviews for the restaurants in this segment revealed two topics, "Bread and dips" and "Dining experience," as important drivers of customer satisfaction. Our analysis also indicated that words such as "bread," "olive," and "sauce" frequently appeared in negative reviews, suggesting Pizzeria Bianco should address concerns related to these aspects to enhance customer satisfaction more effectively (see Table F-2). In contrast to reviewers in the other segments, those within Segment 1 exhibited a lower frequency of Yelp reviews (mean of 83 reviews) and assigned lower ratings (mean of 3.34). Their reviews garnered fewer acknowledgments for usefulness, with an average of 128, compared to averages of 289 in Segment 2 and 265 in Segment 3.

Second, Segment 2, labeled as "Medium satisfaction segment," contained 625 customers (31%) who showed moderate satisfaction ratings (3.39). Customers in this segment discussed topics of "Service performance" and "Food quality and pairing" as determining factors for their satisfaction. Notably, the word "staff" frequently appeared in negative reviews, suggesting an area for potential improvement in service quality. The words "pie" and "oil" were also frequently mentioned in negative reviews concerning the food quality aspects of this restaurant. On average, reviewers in this segment wrote 131 reviews and assigned a mean rating of 3.68 to restaurants on Yelp. In addition, reviewers in this segment received a higher count of "usefulness" compliments (289, compared to128 in Segment 1 and 265 in Segment 3), indicative of a greater level of appreciation for their reviews from others.

Last, Segment 3, referred to as "Highly satisfied customer segment," consisted of 1,105 customers (54%) who expressed very high satisfaction, garnering an average rating of 4.71. Customers in this segment frequently discussed "Pizza quality" and "Expectation management"

as important factors influencing their satisfaction for this restaurant. We noticed that terms such as “expect” and “better” were often found in negative reviews. Given the elevated satisfaction level of customers in this segment, it becomes important for the establishment to effectively manage customer expectations and continue implementing quality control measures for their pizza. On average, reviewers in this segment contributed 154 reviews and assigned a mean rating of 3.93 to restaurants on Yelp, which portrays customers in this segment as both active and generous reviewers. Interestingly, accounting for the distinct reviewer characteristics across segments, we identified heterogeneous key topics related to their ratings and varying levels of satisfaction among the derived segments.

--- Insert Table F-4 about here ---

References

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Table F-1. Model selection for customer-level yelp review data using Chib's (1995) method (number of segments K and tunning parameter λ)

	λ	K=1	K=2	K=3	K=4	K=5
Log Marginal Likelihood	0.1	-3511.1	-2400.2	-1068.4	-1321.7	-3249.7
	0.05	-3485.0	-1648.5	-642.9	-1190.7	-2818.5
	0.01	-3296.5	-1555.2	-280.1*	-667.1	-1879.0
	0.005	-3363.8	-1569.9	-388.9	-713.8	-2179.6
	0	-3391.2	-2512.9	-832.4	-1044.3	-2577.1

Note: The largest value with * indicates the best solution for this data set.

Table F-2. Customer segments and key satisfaction topics for each segment

	Topic	Selected group words	Coefficients of the words (posterior means)
Segment 1 (15%)	Bread and dips	"bread" "oil" "olive" "delicious" "love"	(-0.3, 0.65, -0.58, 0.86, 0.21)
	Dining experience	"enjoy" "experience" "sauce" "staff" "must" "sausage" "amazing" "spot"	(0.26, 0.67, -0.42, -0.49, -0.44, 0.49, 0.92, 0.5)
Segment 2 (31%)	Service performance	"staff" "expect" "wise" "favorite" "amazing"	(-0.26, -0.3, 0.55, 0.51, 0.54)
	Food quality and pairing	"pie" "perfect" "drink" "delicious" "something" "oil" "olive" "sure"	(-0.44, 0.41, 0.3, 0.47, -0.54, -0.69, 0.37, 0.4)
Segment 3 (54%)	Pizza quality	"wiseguy (a popular pizza name in the restaurant)" "better" "differ" "seem" "quit"	(0.15, -0.16, 0.17, -0.27, -0.21)
	Expectation management	"expect" "though" "give"	(-0.2, -0.23, -0.3)

Table F-3. The regression coefficients and significant group words (with a same color) of three customer segments of a restaurant selected by the proposed model with Yelp review data. A total of 81 words (average 27 words per segment; 11.6% of all 233 words) were selected across three derived segments out of 233 words.

Segment 1 (15%)		Segment 2 (31%)		Segment 3 (54%)	
Words	Coeffs	Words	Coeffs	Words	Coeffs
Intercept	1.72	Intercept	3.75	Intercept	4.91
amazing	**0.92**	**amazing**	**0.54**	**better**	**-0.16**
best	0.6	better	-0.32	came	-0.19
bread	**-0.3**	**brick**	**-0.57**	**differ**	**0.17**
call	-0.28	crust	-0.15	**expect**	**-0.2**
delicious	**0.86**	**delicious**	**0.47**	**fresh**	**0.09**
downtown	**0.72**	dinner	0.48	**give**	**-0.3**
enjoy	**0.26**	**drink**	**0.3**	good	-0.14
everi	0.67	enjoy	0.16	much	-0.12
experience	**0.67**	even	-0.32	pretti	-0.43
flavor	-0.25	everyth	0.22	**quit**	**-0.21**
get	-0.24	**expect**	**-0.3**	see	-0.18
glass	0.49	**favorit**	**0.51**	**seem**	**-0.27**
good	0.22	give	-0.33	still	0.13
kind	0.51	great	0.2	**though**	**-0.23**
long	0.45	hype	-0.31	**wiseguy**	**0.15**
love	**0.21**	ingredi	0.37		
made	0.29	**local**	**0.34**		
margherita	0.52	**lot**	**-0.39**		
must	**-0.44**	love	0.13		
oil	**0.65**	much	-0.23		
oliv	**-0.58**	never	-0.38		
said	-0.21	**oil**	**-0.69**		
sauc	**-0.42**	**oliv**	**0.37**		
sausag	**0.49**	**perfect**	**0.41**		
spot	**0.5**	**pie**	**-0.44**		
staff	**-0.49**	rosa	0.24		
star	0.28	sausag	-0.21		
stop	**0.78**	see	0.21		
time	-0.25	**seem**	**0.27**		
		servic	-0.24		
		sit	-0.49		
		someth	**-0.54**		
		staff	**-0.26**		
		sure	**0.4**		

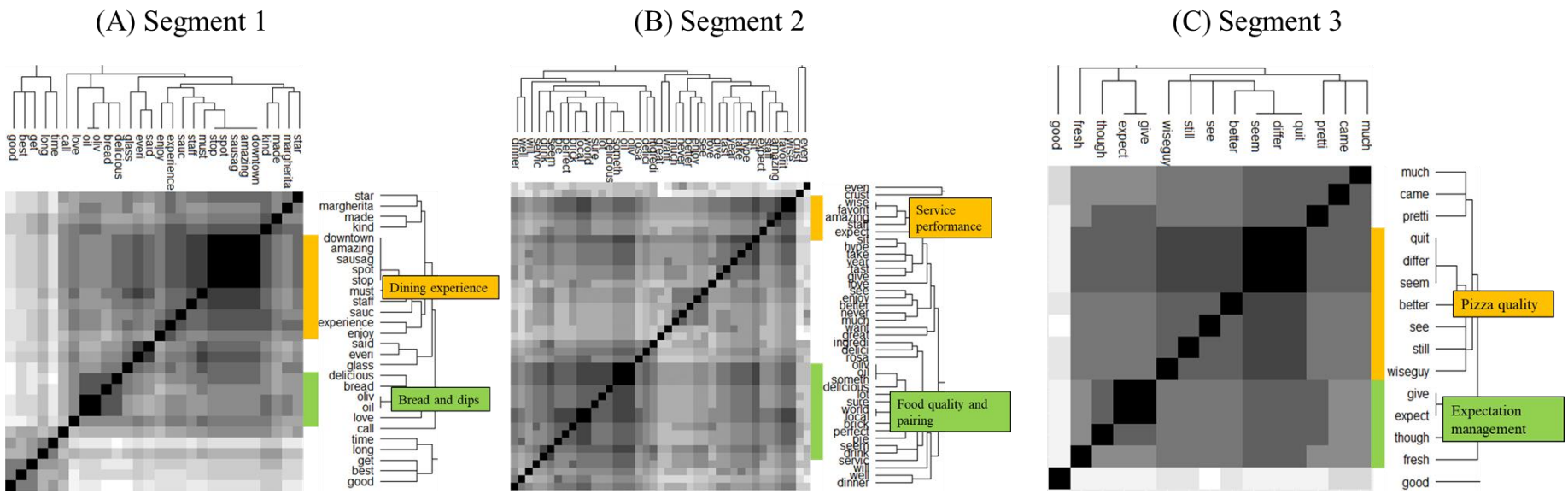
take	-0.31
tast	-0.22
want	-0.31
well	0.55
will	0.22
wise	**0.55**
world	**0.43**
year	-0.29

Note: all shaded cells refer to selected words based on group variable selection; cells pertaining to a particular topic are shaded with orange or green; gray-shaded cells are selected as individual words.

Table F-4. Description of each customer segment of the restaurant.

		Segment 1 "Unhappy customer segment"	**Segment 2** "Medium satisfied customer segment"	**Segment 3** "Highly satisfied customer segment"
Average star rating (satisfaction) about Pizzeria Bianco experience		2.05	3.39	4.71
Numbers of reviewers/customers (%)		306 (15%)	625 (31%)	1,105 (54%)
Key topics in reviews		Brand and dips, Dining experience	Food and pairing, Service performance	Pizza quality, Expectation management
Important areas of improvement		Bread, Olive, Sauce	Pie, Staff	Expect
Profiles of reviewers in Yelp	Average number of reviews by reviewers	83	131	154
	Mean of average ratings by reviewers to restaurants on Yelp	3.34	3.68	3.93
	Mean number of 'usefulness' that reviewers received	128	289	265

Figure F-1. Hierarchical clustering maps (heatmaps) to identify topics of three customer segments of the restaurant (see Table F-2 for selected group words for each topic).



Note: Black-shaded portions of each heatmap show the pattern of groups of words in this hierarchical clustering (topics) that significantly affect the star rating. For the first panel of segment 1, words next to bars of two different colors belong to topics of "Dining experience" and "Bread and dips." For the second panel of segment 2, words next to two different colors belong to topics of "Service performance" and "Food quality and pairing." For the last panel of segment 3, words next to two different colors belong to topics of "Pizza quality" and "Expectation management."

Web Appendix G: Empirical Application Study with Professor Rating Review Data

Background and Data Description

The annual revenue of the college and university industry is $528.4 billion (IBISWorld, 2020). Managing education quality and properly caring for students are important, not only because of the large size of the higher education industry but also because of a responsibility to society. To achieve these goals, colleges and universities regularly check students' opinions through formal, structured teaching evaluation surveys, which include questions about diverse attributes (e.g., class interest, teacher involvement). However, students also talk, share, and complain about their professors on online review sites. Educational institutions should pay attention to such natural voices from students and might need to systematically and periodically summarize patterns from the unstructured review texts.

We collected data from RateMyProfessors.com, on which college and university students post ratings and texts reviews for their professors. This online review site includes ratings for more than 7,500 schools and 1.7 million professors (see "About Rate My Professors," n.d.). On the collected data from RateMyProfessors.com, students provided star rating to the professors with one out of five ratings across helpfulness, easiness, and clarity (i.e., 1 star: lowest score, 5 stars: highest score) and can write textual reviews.

Our data include five-year reviews from 2010 to 2015. In the collected data, the number of reviews can vary depending on the professor. We analyze the data at the level of each rated professor (which includes multiple reviews per professor) as a unit of analysis. We remove any professors with a small number of reviews (i.e., 10 or less) from our dataset. The used data include star ratings (e.g., helpfulness, clarity), information about institutions and disciplines, and

unstructured textual reviews for 16,027 professors with 615,963 reviews from 139 institutions and 14 disciplines. Of the 139 institutions represented, 108 had high research activity (The Carnegie Classification of Institutions of Higher Education, 2016), and all the top 31 liberal arts colleges, according to a U.S. News & World Report ranking (2016) were included. The majority of the data represented professors from three academic categories: Business (22%; e.g., accounting, economics, finance, information systems, management, and marketing), STEM (Science, Technology, Engineering, and Mathematics; 63%; e.g., biology, engineering, mathematics, and statistics), and Social Science (15%; e.g., psychology and sociology).

Model Comparison

Using the same way as we did in the first application study, we compared the proposed model with the same set of six benchmark models. We used helpfulness rating as a dependent variable because helpfulness and clarity are combined as a good indicator of class quality (Abrami, Cohen and d'Apollonia 1988, Stroebe 2020) and the two ratings are highly and positively correlated in this data (ρ = 0.92, p < 0.01). A word frequency matrix used as predictors included 205 words that appeared an average of one time per review across at least 10 professors.

For the proposed model, using log marginal likelihood criteria suggested by Chib (1995), we selected a four-segment model with a 0.01 as a λ tuning parameter value as the best solution, as shown in Table G-1. For comparison, we presented four-segment solutions by latent class regression (flexmix) and latent class regression with variable selection (Kim et al., 2012)

--- Insert Table G-1 about here ---

In the first empirical application study, we discussed model performance and methodological advantages in prediction, interpretability and heterogeneity between the proposed model and benchmark models. Please see discussion about methodological strength/weakness of each benchmark models in the first application study. Avoiding duplications but checking robustness in prediction, we would compare MAE, RHR and coherence score between the proposed model and benchmark models. As shown at Table G-2, the unsupervised topic model showed the worst predictive ability out of the seven tested models (MAE = 0.592 and RHR = 74%), and the hierarchical topic model (Teh et al. 2006) also showed the somewhat low predictive performance among the tested models (MAE = 0.583 and RHR = 74.4%). OLS using the words frequency matrix of 205 words slightly improved prediction by the topic model with 0.584 of MAE and 74.4% of RHR in this data. Supervised topic model improved MAE and RHR to 0.514 and 78.5%, but its predictive performance was poorer than other supervised mixture models accommodating segment-level heterogeneity. As traditional latent class regressions, flexmix *without* variable selection showed better prediction than the two topic models and OLS with 0.488 of MAE and 81.8% of RHR. Next, Kim et al.'s model (2012) *with* variable selection improved prediction performances from flexmix by providing a reduced MAE (= 0.447) and an increased RHR (= 82.9%). Finally, the proposed model showed a best prediction performance among the six tested models with a MAE (= 0.445) and a RHR (= 83%), which was slightly better than the Kim et al.'s model. Additionally, we tested two more benchmarks. We ran traditional topic model to find topics and then employed (i) regression and (ii) flexmix with the topics and their two-way interactions. But their prediction performances were worse than the proposed model's prediction as presented in the Table G-2.

To evaluate models' interpretability, we compared the coherence scores among the seven tested models, which is a mathematical measure related to human interpretability (Zvornicanin 2022). As presented in Table G-2, the coherence score of the proposed model is closest to zero ($= -0.872$), which means higher coherence (interpretability) than any other six benchmark models. As expected, three supervised regression models (aggregate regression, flexmix, and Kim et al.'s (2012)) without considering any dependence between words show the three worst coherence scores of -1.214, -1.237 and -1.271, respectively. Next, three topic models (unsupervised topic model, hierarchical topic model, and supervised topic model) showed better coherence scores (-1.007, -0.987 and -1.032, repectively) but their coherences were still worse than the score by the proposed model.

As an example for interpretability issue with the supervised regression models, Kim et al.'s model (2012) selected more than 20 words, including "horrible," "amazing," "confusing," "care," "difficult," "great," "help," and "lecture" in Segment 4 (see Table G-3). But, since their model did not reveal any group structure among the selected words, managers must analyze each word individually in relation to the star rating, which can lead to difficulties in interpreting the results. In contrast, the proposed model selected groups of co-occurring words for describing topics with selected groups of neighbor words across segments. For example, as presented at Table G-4, the first topic, "Instructor," in Segment 1 was described with a group of words: "funny," "instructor," "interesting," and "passion." Another topic, "Grading," in Segment 2 was associated with a group of words: "grade," "extra," "point," and "drop."

--- Insert Tables G-2 and G-3 about here ---

In sum, the proposed model showed best performance in terms of predictability and interpretability among the seven compared models. Finally, Table G-4 also shows that the

proposed model provides segment-wise topics for accommodating heterogeneity, which is not possible using aggregate-level models (e.g., OLS and various topic models).

--- Insert Table G-4 about here ---

Segment-Specific Results

Using the proposed model, we determined four segments, each with several groups of words (i.e., topics). As presented in Table G-4, each segment included multiple topics with label of each topic, and each topic contained co-occurring relevant words. As evidenced by the estimated coefficients from Table G-4, we observed good face validity. For example, we found that frequent appearances of positive words, such as "interesting," "great," and "easy" were positively correlated with higher star ratings (i.e., positive coefficients of the positive words). Note, Table G-5 provides regression coefficients of all selected words across four segments. Similarly, frequent appearance of negative words, such as "tough," "confuse," and "difficult" were negatively correlated with lower star ratings (i.e., negative coefficients of negative words). As discussed in the first application study, neutral words (e.g., "class," "professor," or "lecture") could be correlated in either direction by accommodating their frequency pattern between positive reviews and negative reviews. As shown Figure G-1, we employed heatmap, to elucidate the graphical patterns of groups amongst the selected words, which can be described as topics.

--- Insert Table G-5 and Figure G-1 about here ---

Segment 1 (the most dissatisfied segment) was a relatively small segment, with 21.5% (3,451 professors) of the data. We found that this segment provided significantly lower mean of helpfulness rating (2.73 vs. 3.41 of Segment 2, 3.96 of Segment 3, 4.39 of Segment 4; $p<0.05$)

compared to other three segments (see Table G-6). Students in Segment 1 talked two topics related to their ratings in their reviews. The first topic was described as “Clarity” with selected group words of “clear,” “review,” “fine,” and “succeed,” with positive coefficients. The second topic was described as “Instructor” with selected group words of “instructor,” “passion,” “funny” and “interesting.” With a profiling analysis (Table G-6), this dissatisfied segment can be characterized as more STEM-oriented segment, given a higher proportion of segment 1 came from STEM majors. As representative example disciplines, this segment included higher proportions from computer science, economics, and statistics than other segments. Students in this segment care more about clarity of course materials as well as instructors.

Segment 2 (the second dissatisfied segment) was also a small segment with 23% (3,690 professors) of the data. As a first topic, similar to a topic of the segment 1, we would describe “Clarity” based on the selected group words of “clear,” “wish,” “confuse,” and “worst.” As a second topic, we found “Grading” as evidenced with words of “grade,” “extra,” “point,” and “drop.” This segment can be characterized as a more social science oriented segment (e.g., sociology). The result suggests that professors in this segment should care about “Grading” and “Clarity of course materials.”

Segment 3 (the second satisfied segment) was 23.8% (3,809 professors) of the data, and we identified two topics related to the dependent rating. First, we described “Care” from the selected group words of “care,” “tough,” “confuse,” and “teach.” Second, we described a topic of “Fun” based on the group words of “fun,” “interest,” “amazing,” and “great,” where all coefficients are positive. This segment could be characterized as a segment related to business studies. As representative example disciplines, this segment included higher proportions from finance and management than other segments.

Segment 4 was the most satisfied, largest segment with 31.7% (5,077 professors) of the data. We found only major concern of this segment was “Difficulty” as evidenced with the group words of “avoid,” “difficult,” and “hard,” where all coefficients are negative. We found that the Segments 3 and 4 (more satisfied groups) could be characterized as business and social science disciplines. Especially, this segment included higher proportions from psychology than other segments.

From the empirical study 2, we consistently found the proposed model not only outperforms the benchmark models in predicting star ratings but also helps marketers understand segment-specific interpretable key drivers of star ratings by identifying groups of words.

--- Insert Table G-6 about here ---

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Table G-1. Model selection (i.e., the number of segments [K] and λ) for the proposed model with the professor rating review data, using Chib's (1995) method.

	λ	K=1	K=2	K=3	K=4	K=5
Log Marginal Likelihood	0.1	-17877.9	-17096.1	-16609.4	-16194.9	-17320
	0.05	-17802.5	-17066.8	-16541.8	-16101.5	-17301.9
	0.01	-17736.1	-16935.2	-16494	-16055.1*	-17285.4
	0.005	-17795.9	-17011.3	-16522.5	-16098	-17302.7
	0	-17895.7	-17104.6	-16606.5	-16220.4	-17328.6

Note: The largest value with * indicate the best solution for this data set.

Table G-2. Predictive ability comparison with professor rating review data between the proposed model and benchmark models.

Learning Type	Unsupervised				Supervised				
Models	Topic Model Regression	Hierarchical Topic Model (HDP) Regression	Topic Model Regression *with* Interactions	Topics Model Latent Class regression *with* Interactions	Supervised Topic Model (McAuliffe and Blei 2010)	Aggregate Regression (OLS) [a]	Latent Class Regression *without* Variable Selection [a]	Latent Class Regression *with* Word Selection (Kim et al., 2012) [a]	**The Proposed Model with Topic (grouped words) Selection** [a]
Error for Prediction (MAE)	0.592	0.583	0.550	0.488	0.514	0.584	0.488	0.447	**0.445**
Recommendation Hit Rate (RHR)	74%	74.4%	76.4%	81.8%	78.5%	74.4%	81.8%	82.9%	**83%**
Coherence Score (with UMass)	-1.007	-0.987	-	-	-1.032	-1.214	-1.237	-1.271	**-0.872**

Note, for MAE, lower value is better; for RHR higher rate value is better; for Coherence Score, a value to closer to zero means high coherence (i.e., better interpretability); coherence score is not available for the models with interaction terms.

[a] All the estimated coefficients and their significances are available from the first author.

Table G-3. Benchmark result by latent class regression *with* variable selection (Kim et al. 2012) with significantly selected 63 words out of 205 words with Professor review data.

Segment 1 (2%)		Segment 2 (7%)		Segment 3 (47%)				Segment 4 (44%)	
Words	Coeffs	Words	Coeffs	Words	Coeffs	Words	Coeffs	Words	Coeffs
awesom	0.23	question	-0.04	class	-0.01	helpful	0.07	lectur	-0.02
worst	0.01	fair	0.06	lectur	-0.03	respect	0.11	help	0.01
fair	0.09	also	0.08	help	0.03	anyth	-0.09	hard	-0.03
		keep	0.13	will	0.02	sure	0.04	read	-0.03
				hard	-0.04	tell	-0.04	tough	-0.03
				read	-0.04	avoid	-0.34	best	0.05
				clear	0.05	fun	0.06	great	0.02
				best	0.13	excel	0.09	difficult	-0.04
				ask	-0.06	confus	-0.15	worst	-0.77
				great	0.1	seem	-0.07	amaz	0.04
				grade	-0.03	never	-0.04	just	-0.02
				realli	0.03	favorit	0.07	care	0.03
				easi	0.05	say	-0.05	ever	0.02
				interest	0.05	talk	-0.06	made	0.03
				expect	-0.05	bad	-0.08	avoid	-0.18
				amaz	0.14	high	0.06	confus	-0.05
				know	-0.03	stay	-0.08	fast	-0.03
				time	-0.02	awesome	0.16	talk	-0.03
				good	0.03	hours	0.05	interesting	0.03
				teach	-0.06	word	-0.04	peopl	-0.06
				point	-0.06	wish	0.1	passion	0.04
				love	0.09	horribl	-0.23	better	-0.04
				alway	0.03			doesnt	-0.09
								horribl	-0.09

Table G-4. Topics based on significantly selected group words per segment by the proposed model with Professor review data.

	Topics	Selected group words	Regression coefficients of the words (posterior means)
Segment 1 (21.5%)	Clarity	"clear" "review" "fine" "succeed"	(0.06, 0.07, 0.07, 0.16)
	Instructor	"funny" "instructor" "interesting" "passion"	(0.11, 0.09, 0.11, 0.14)
Segment 2 (23%)	Clarity	"clear" "wish" "confuse" "worst"	(0.07, 0.12, -0.19, -0.1)
	Grading	"grade" "extra" "point" "drop"	(-0.04, 0.07, -0.08, -0.08)
Segment 3 (23.8%)	Care	"care" "tough" "confuse" "teach"	(0.06, -0.06, -0.11, -0.04)
	Fun	"fun" "interest" "amaze" "great"	(0.06, 0.05, 0.09, 0.04)
Segment 4 (31.7%)	Difficulty	"avoid" "difficult" "hard"	(-0.14, -0.04, -0.02)

Table G-5. The regression coefficients and selected group words (with a same color) of four segments (K=4) selected by the proposed model with Professor rating data. A total of 70 words (average 28 words per segment; 13% of total 205 words) were selected across four segments.

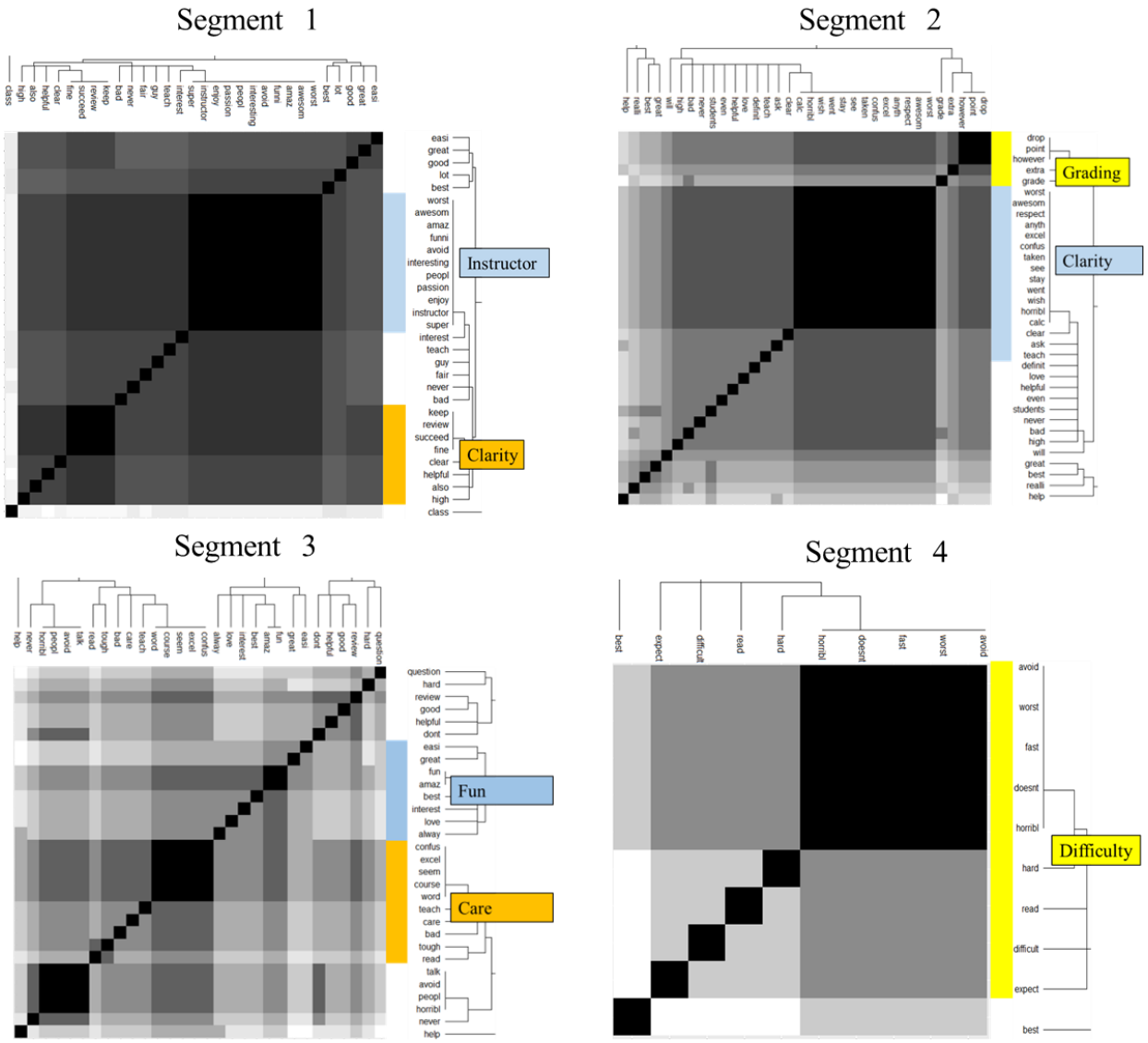
Words	k=1	k=2	k=3	k=4	Words	k=1	k=2	k=3	k=4
(Intercept)	2.2	3.34	4.13	4.44	realli	0	0.04	0	0
class	-0.01	0	0	0	extra	0	0.07	0	0
clear	0.06	0.07	0.02	0	point	0	-0.08	0	0
best	0.13	0.14	0.05	0.04	definit	0	0.07	0	0
awesom	0.31	0.18	0.01	0	love	0	0.1	0.05	0
great	0.08	0.12	0.04	0	respect	0	0.13	0	0
easi	0.07	0	0.04	0	drop	0	-0.08	0	0
lot	0.05	0	0	0	anyth	0	-0.1	0	0
worst	-0.14	-0.1	0	-1.12	even	0	-0.06	0	0
interest	0.06	0	0.05	0	however	0	-0.09	0	0
amaz	0.47	0	0.09	0	excel	0	0.1	0.07	0
good	0.03	0	0.03	0	confus	0	-0.19	-0.11	0
teach	-0.04	-0.06	-0.04	0	students	0	-0.11	0	0
guy	-0.06	0	0	0	taken	0	-0.13	0	0
helpful	0.08	0.08	0.06	0	see	0	-0.15	0	0
fair	0.1	0	0	0	stay	0	-0.12	0	0
funni	0.11	0	0	0	went	0	-0.07	0	0
avoid	-0.1	0	-0.6	-0.14	wish	0	0.12	0	0
also	0.05	0	0	0	horribl	0	-0.29	-0.12	-0.05
review	0.07	0	-0.07	0	calc	0	0.09	0	0
never	-0.06	-0.05	-0.04	0	hard	0	0	-0.04	-0.02
bad	-0.06	-0.08	-0.09	0	read	0	0	-0.05	-0.03
interesting	0.11	0	0.06	0	tough	0	0	-0.06	0
peopl	-0.06	0	-0.11	0	question	0	0	-0.04	0
passion	0.14	0	0	0	alway	0	0	0.04	0
high	0.07	0.08	0	0	care	0	0	0.06	0
enjoy	0.1	0	0	0	fun	0	0	0.06	0
keep	0.13	0	0	0	talk	0	0	-0.08	0
instructor	0.09	0	0	0	ask	0	-0.07	0	0
succeed	0.16	0	0	0	course	0	0	-0.11	0
anyon	-0.1	0	0	0	word	0	0	-0.15	0
fine	0.07	0	0	0	difficult	0	0	0	-0.04
super	0.11	0	0	0	expect	0	0	0	-0.05
help	0.01	0.03	0.02	0	fast	0	0	0	-0.06
will	0	0.03	0	0	doesnt	0	0	0	-0.14
grade	0	-0.04	0	0	**Mixture**	**21.5%**	**23.0%**	**23.8%**	**31.7%**

Note: all shaded cells refer to selected words based on group variable selection; cells pertaining to a particular topic are shaded yellow, orange, or blue; gray-shaded cells are selected as individual words.

Table G-6. Description of each segment for the Professors sample.

	Segment 1	**Segment 2**	**Segment 3**	**Segment 4**
Mean helpfulness	2.73	3.41	3.96	4.39
Proportion	21.5%	23%	23.8%	31.7%
N of Professors	3,451	3,690	3,809	5,077
Category	STEM	Social Science	Business	Social Science/Business
Discipline characteristics	Computer science, Economics, Statistics	Sociology	Finance, Management	Psychology
Key topics in reviews	Clarity, Instructor	Clarity, Grading	Care, Fun	Difficulty

Figure G-1. Hierarchical clustering maps to identify topics of four segments for Professor rating review data.



Note: Black-shaded portions of each heatmap show the pattern of groups of words with smaller distances in this hierarchical clustering (potential topics) that significantly affect the helpfulness rating. For the top left panel of segment 1, words next to bars of two different colors belong to topics of "Instructor" and "Clarity." For the top right panel of segment 2, words next to bars of two different colors belong to topics of "Grading" and "Clarity." For the bottom left panel of segment 3, words next to bars of two different colors belong to topics of "Fun" and "Care." For the bottom right panel of segment 4, we found one topic of "Difficulty."

Web Appendix H: Sample Trace Plots to Check Convergence by the Proposed Model

Figure H-1. Sample trace plots for posterior MCMC after burn-in to check convergence (Yelp).

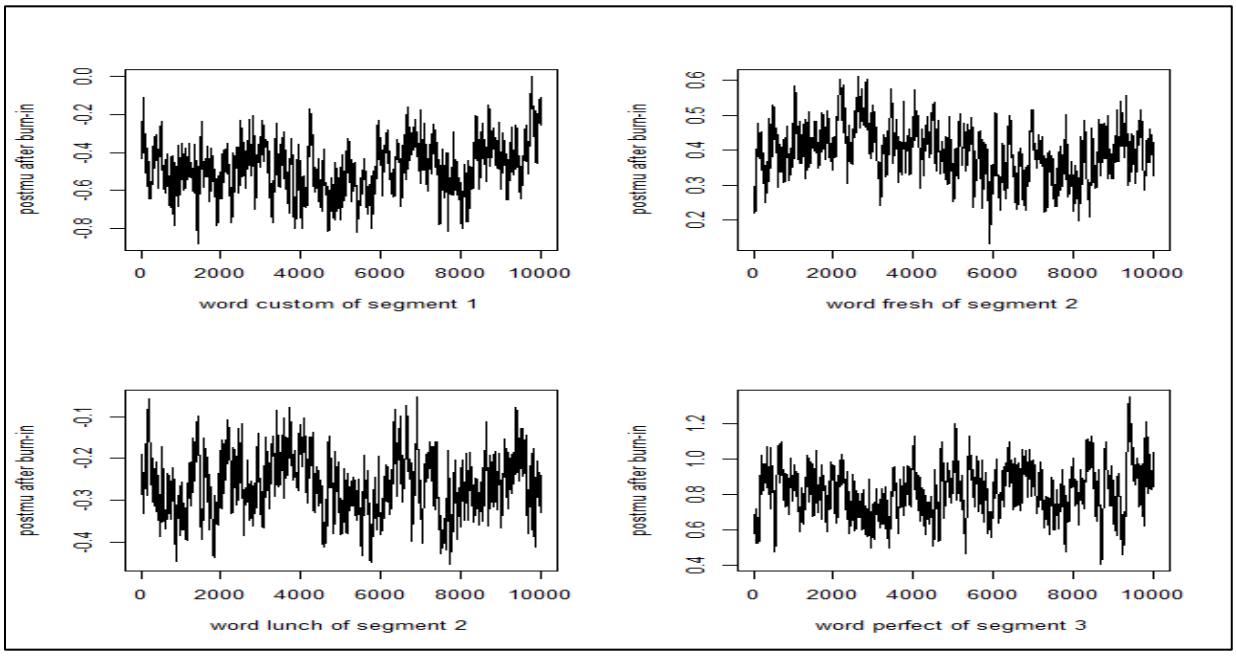
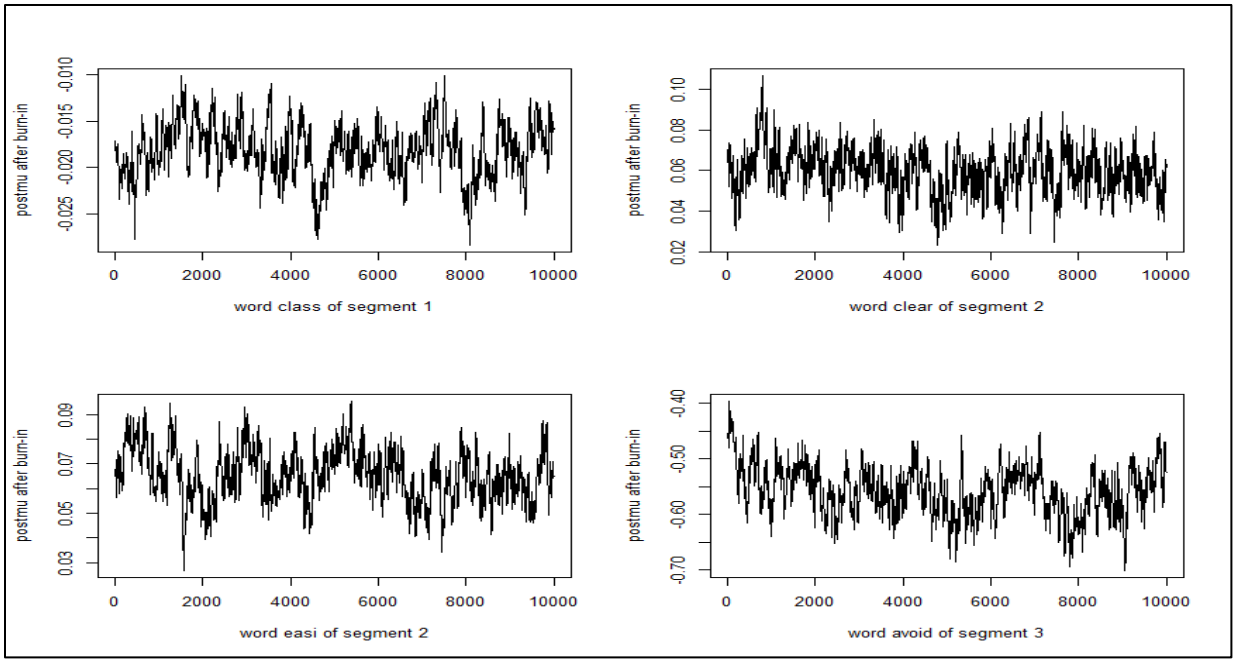


Figure H-2. Sample trace plots for posterior MCMC after burn-in to check convergence (Professor rating).



Web Appendix I: Survey Material for Comparing Text Analytics Methods

Dear MS Business Analytics Student:

[Consent form for study participants here]

(Q1) Are you a MS Business Analytics student?

(1) Yes (2) No

(Q2) Age? _____

Imagine that you are a data analytics consultant. You wanted to analyze online textual reviews from Yelp to better understand customer voices on restaurants. More specifically, your goal is to figure out important factors that affect customer satisfaction on restaurants. To this end, two different text analytics methods were used, providing groups of relevant (i.e., frequently co-occurring) words. Note, the groups of words identified by each method need to be described (or labeled) as understandable factors (topics/dimensions/attributes) for customer satisfaction.

For example, in Mejia et al., 2021

- the set of words **{order, server, came, ask, minute}** was labeled as "**Responsiveness"**
- the set of words **{good, dish, flavor, chicken, sauce}** was labeled as "**Food quality"**

In the next two pages, you will be shown groups of words identified by each text analytics method (method A and B) one at a time.

Reference

Mejia, Jorge, Shawn Mankad, and Anandasivam Gopal (2021) "Service quality using text mining: Measurement and consequences," *Manufacturing & Service Operations Management*, 23 (6), 1354-1372.

Note: Presentation order of methods A and B was randomized during survey data collection

Method A probabilistically identifies groups of frequently co-occurring words from textual reviews, which later needs to be labeled as meaningful topics for customer satisfaction. It also divides samples into segments that share common drivers for customer satisfaction. Method A was applied to the Yelp customer reviews dataset. As a result, it found 3 segments of restaurants and 3~4 groups of frequently co-occurring words for each segment.

(Q3-1) Please take a look at the result table below and try to label each topic based on the set of frequently co-occurring words. If it is difficult to find a reasonable label for a given set of words, you may leave it blank and skip to the next one.

Segment	Topic	Set of Co-occurring Words	Label
Segment 1	**Topic 1**	{kid, dessert, fun, help, park, always}	
	Topic 2	{best, meal, eat, taste, try, also}	
	Topic 3	{night, bar, live, love, close}	
	Topic 4	{service, minute, ask, never, got, even}	
Segment 2	**Topic 1**	{great, friend, love, perfect}	
	Topic 2	{eat, taste, fresh, try, cook, fish, serve}	
	Topic 3	{location, park, patio, enjoy, well, open}	
Segment 3	**Topic 1**	{dine, best, server, sauce, salsa, chicken, call, really, can, kid}	
	Topic 2	{manager, service, ask, wait, walk, came}	
	Topic 3	{waitress, bad, never, drink}	

(Q3-2) How much do you agree with the following statements? (1: strongly disagree; 7 strongly agree)

	1 strongly disagree	2 disagree	3 somewhat disagree	4 neither agree nor disagree	5 somewhat agree	6 agree	7 strongly agree
It was easy to label each topic given a set of words identified by the model.							
The sets of words identified by the model represent important factors (attributes) for customers satisfaction on restaurants.							
The results provide useful information for restaurant managers.							

Method B probabilistically finds groups of words based on their co-occurrence in textual reviews, which later needs to be labeled as meaningful topics that form the basis of customer satisfaction. Method B was applied to the Yelp customer reviews dataset. As a result, it found 10 groups of frequently co-occurring words.

(Q4-1) Please take a look at the result table below and try to label each topic based on the set of frequently co-occurring words. If it is difficult to find a reasonable label for a given set of words, you may leave it blank and skip to the next one.

Topic	Set of Co-occurring Words	Label
Topic 1	{drink, place, like, just, food, great, one, get, look, good}	
Topic 2	{always, good, get, love, time, great, like, one, can, just}	
Topic 3	{food, restaurant, menu, meal, service, one, serv, came, just, enjoy}	
Topic 4	{good, order, try, like, really, just, side, got, taste, also}	
Topic 5	{wait, food, service, time, order, minute, came, ask, get, back}	
Topic 6	{place, fresh, lunch, food, price, good, try, clean, like, will}	
Topic 7	{place, best, like, get, just, hot, one, can, make, try}	
Topic 8	{great, place, good, love, friend, back, try, definite, nice, will}	
Topic 9	{food, chicken, good, order, place, lunch, eat, restaurant, like, hot}	
Topic 10	{order, time, location, custom, get, one, ask, just, work, said}	

(Q4-2) How much do you agree with the following statements? (1: strongly disagree; 7 strongly agree)

	1 strongly disagree	2 disagree	3 somewhat disagree	4 neither agree nor disagree	5 somewhat agree	6 agree	7 strongly agree
It was easy to label each topic given a set of words identified by the model.							
The sets of words identified by the model represent important factors (attributes) for customers satisfaction on restaurants.							
The results provide useful information for restaurant managers.							